

Processing sequencing-based spatially-resolved transcriptomic (SRT) data

Åsa Björklund
NBIS, SciLifeLab, Chalmers University of Technology

26-10-06

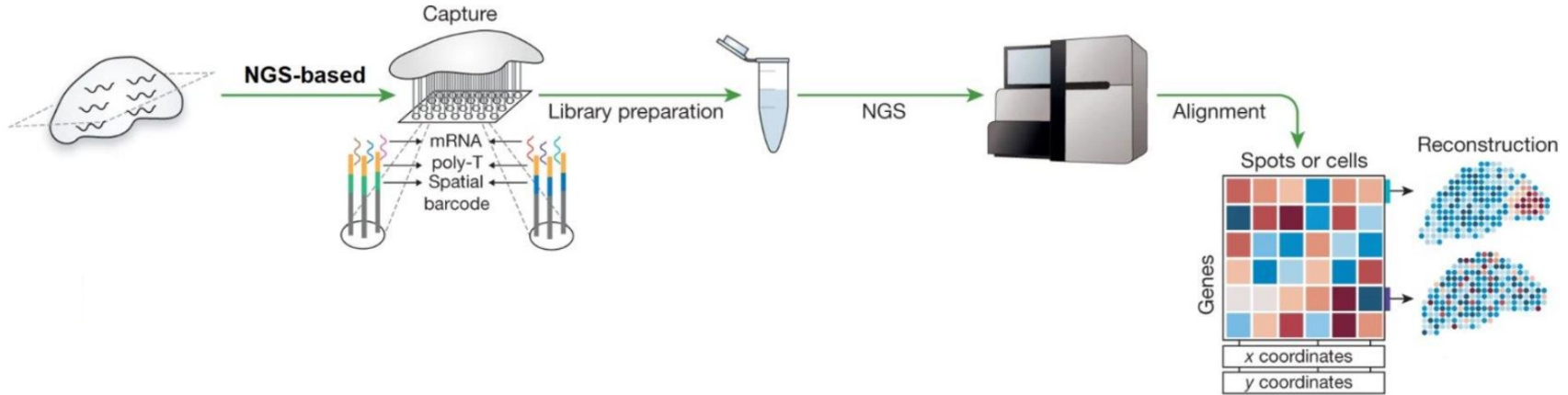
Credit to Ahmed Mahfouz for slides from 2025 course instance

Preprocessing SRT data

Like scRNA-seq data preprocessing:

- Reads to count matrix
- Quality control (QC)
- Normalization
- Feature selection
- Batch correction
- PCA
- UMAP
- Graph clustering
- Differential expression
- Celltype prediction

Sequencing-based SRT



Read alignment (like scRNA-seq):

- SpaceRanger from 10x Genomic
- Stereo-seq Analysis Workflow SAW from STOmics

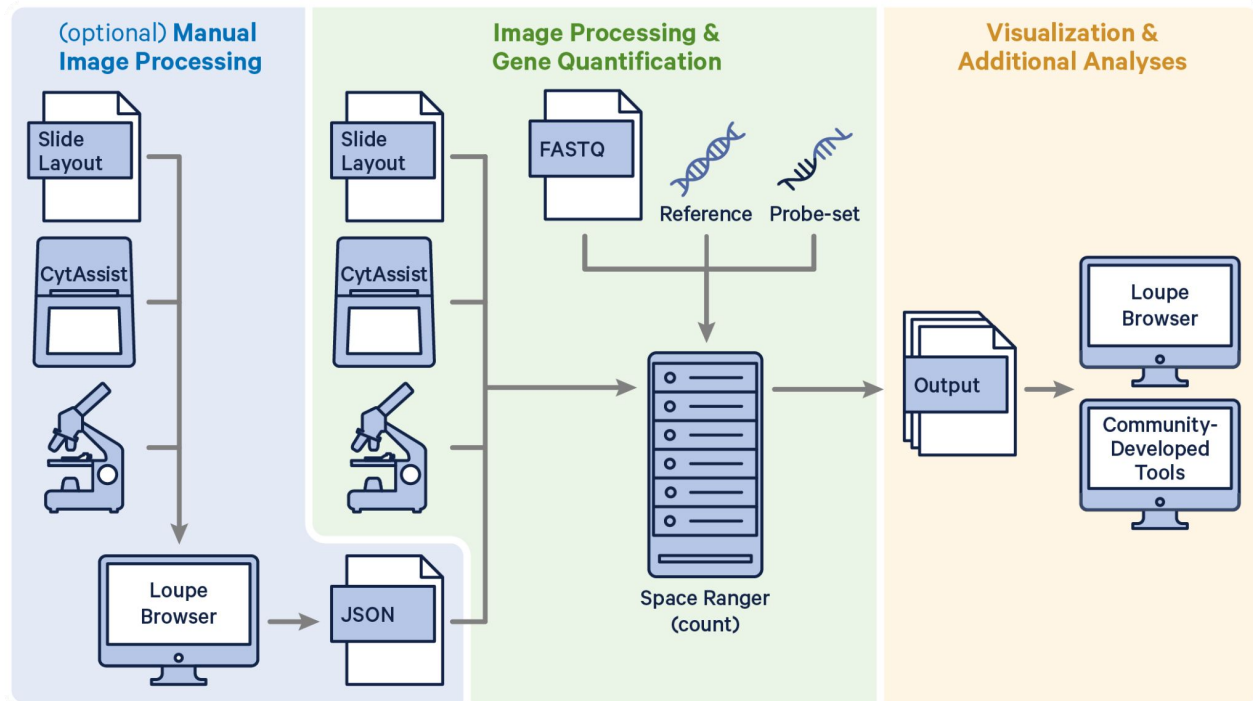
10x SpaceRanger

Similar to 10x Cell Ranger but barcodes are now spatial locations instead of individual droplets.

- Read trimming
- Read mapping (genome or probeset)
- Barcode correction
- UMI correction
- Binning (for VisiumHD)

Image analysis with:

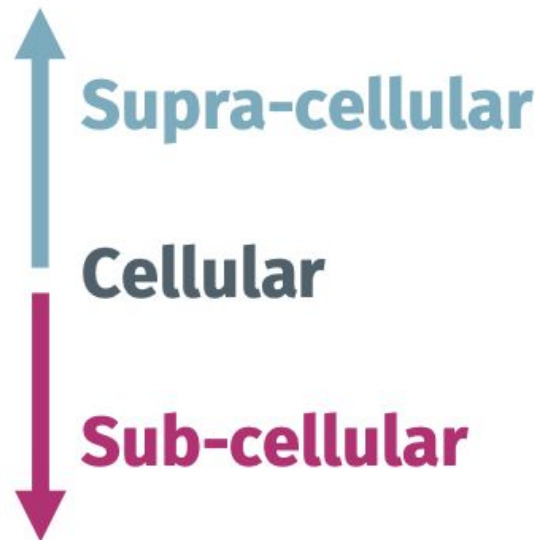
- Fiducial alignment
- Tissue detection
- Image registration
- Nuclei segmentation



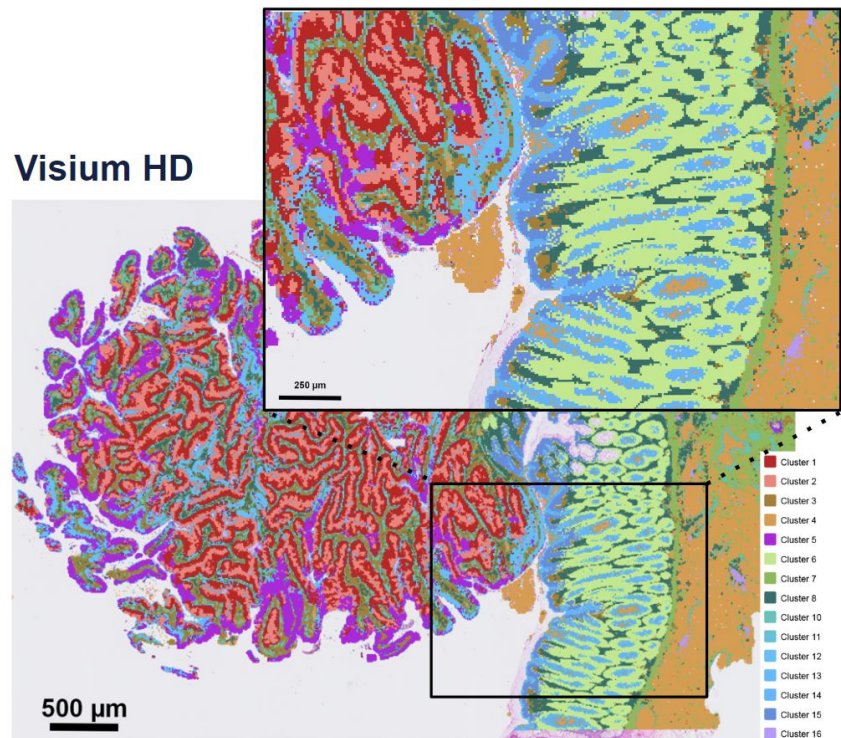
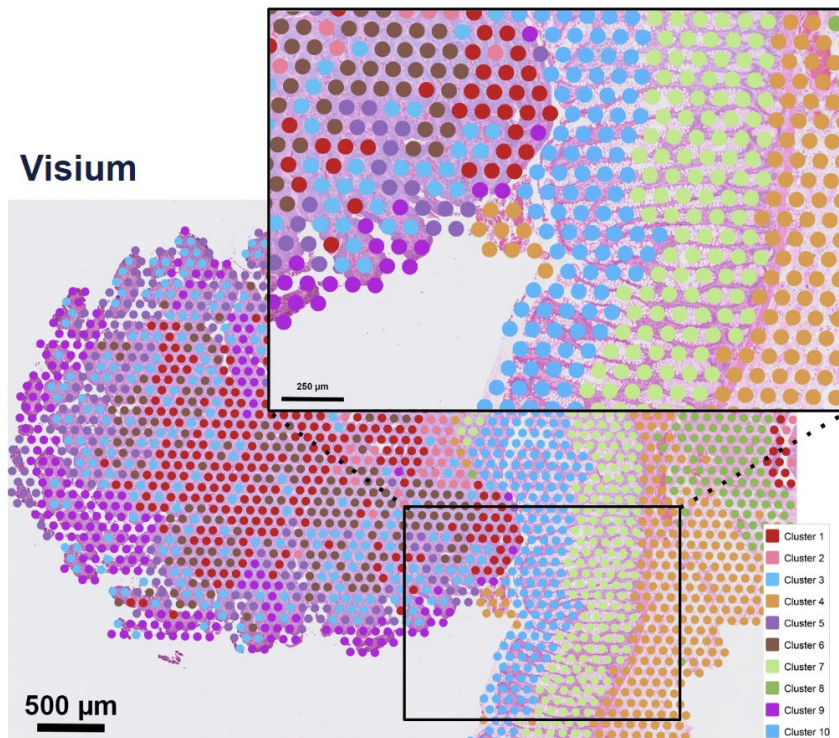
Resolution of seq-based SRT

How large are the “spots?”

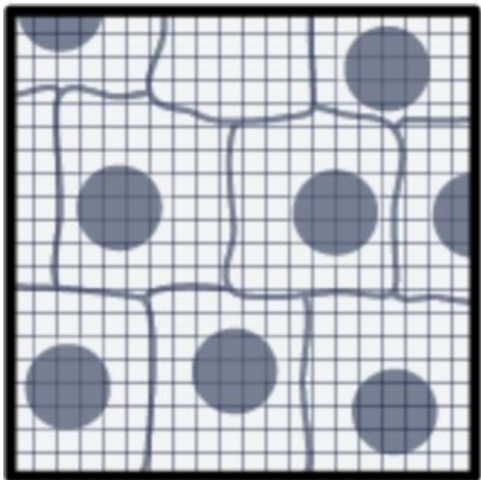
ST	100 μm
Visium	55 μm
DBiT-seq	10-50 μm
Slide-Seq	10 μm
HDST	2 μm
PIXEL-seq	1.22 μm
Seq-Scope	0.5-0.8 μm
Stereo-seq	0.22 μm



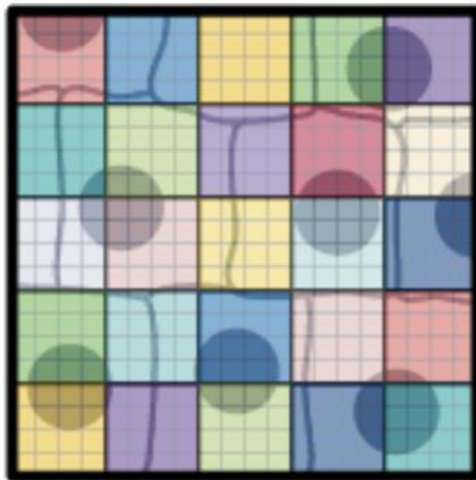
Resolution of seq-based SRT



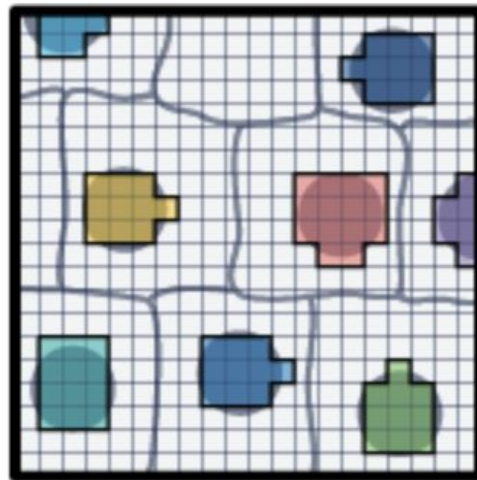
Resolution of seq-based SRT



2x2 μm



8x8 μm



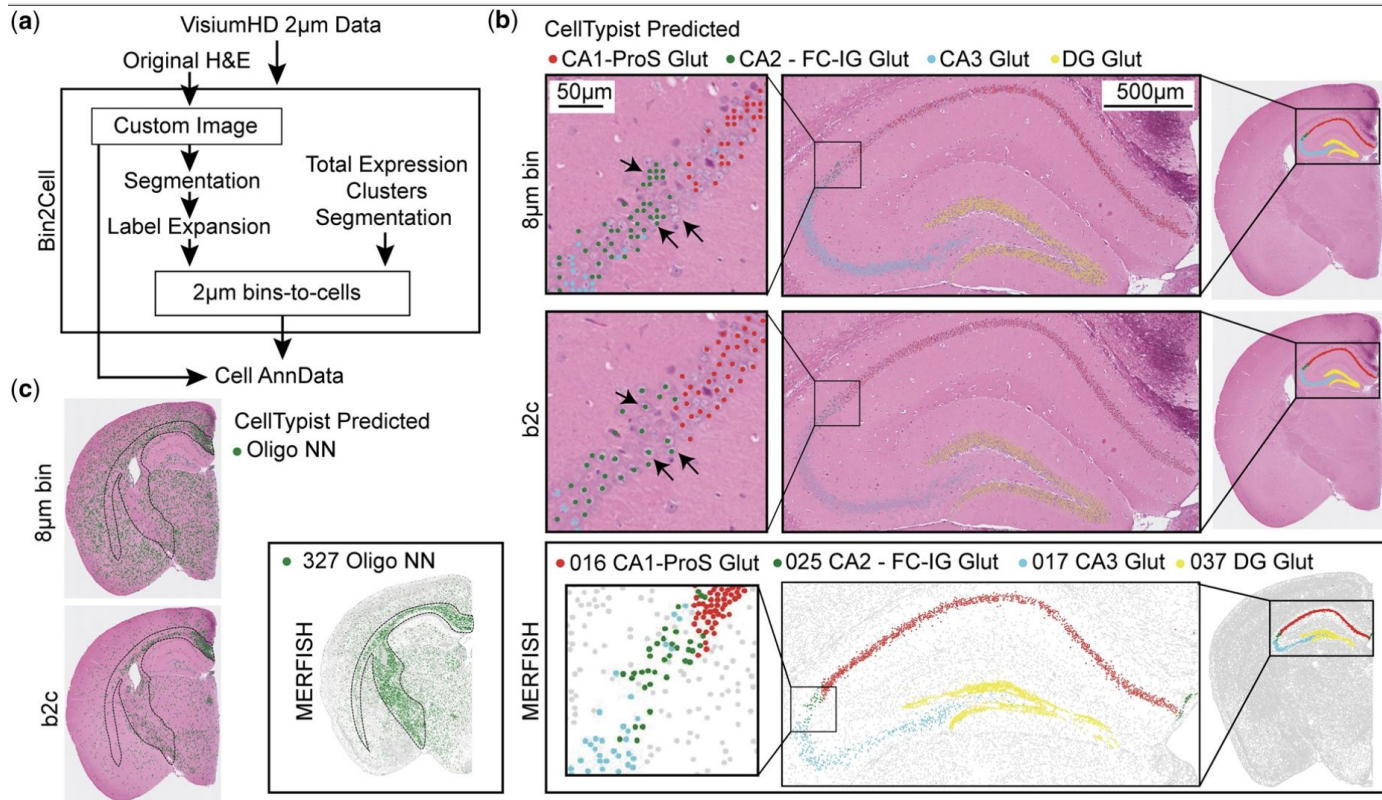
Nuclei Segmentation

From Space Ranger v4.0 VisiumHD output also contains nuclei segmentation and cell segmentation with StarDist.

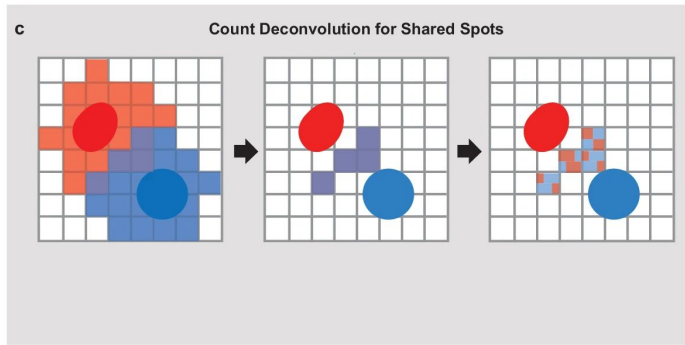
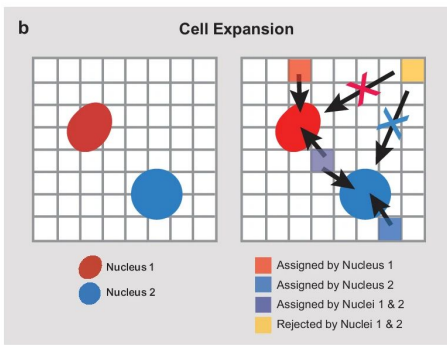
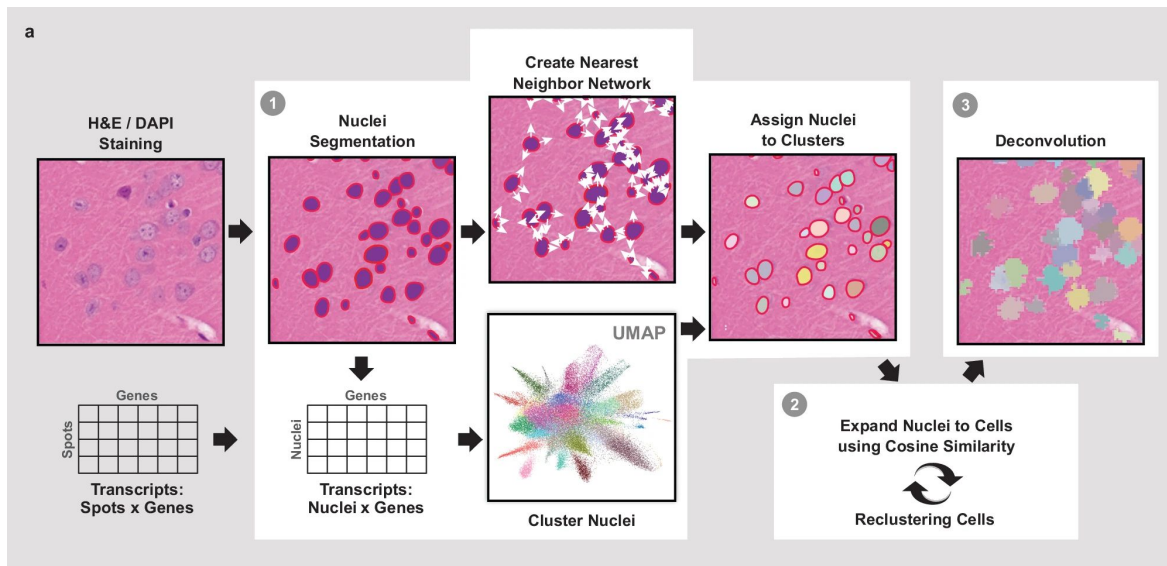
Binning of VisiumHD data

- Default from SpaceRanger 2 μm -> 8 μm -> 16 μm
- Hexagonal bins with SpatialData
- Bin2cell, SMURF etc. - uses nuclei segmentation and transcriptional signature to assign 2 μm regions into “cells”

bin2cell

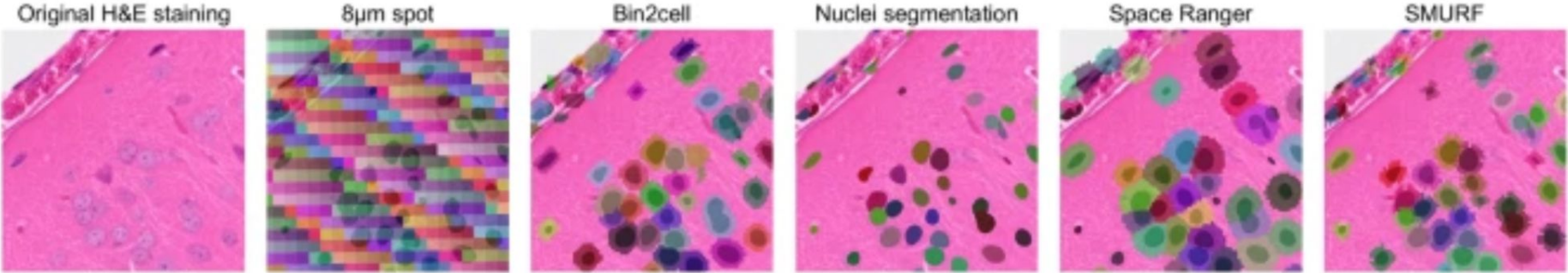


SMURF (Segmentation and Manifold UnRolling Framework)

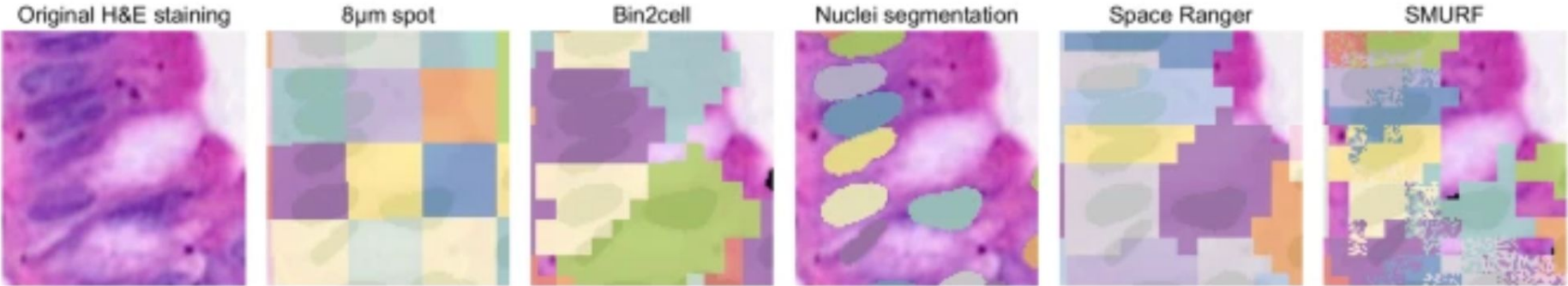


Guo et al. Nature
Comm. 2026

a **Mouse brain**



b **Mouse small intestine**



Quality Control

Factors that affect data quality in scRNA-seq

Cell dissociation

Cell capture

Cell lysis

Reverse transcription

Preamplification

Library preparation and sequencing

Factors that affect data quality in seq-based SRT

Cell dissociation

Cell capture

Cell lysis

Reverse transcription

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Library preparation and sequencing

Factors that affect data quality in seq-based SRT

Cell lysis

Reverse transcription

Library preparation and sequencing

Folds and tears in tissue

RNA diffusion

Uneven sectioning

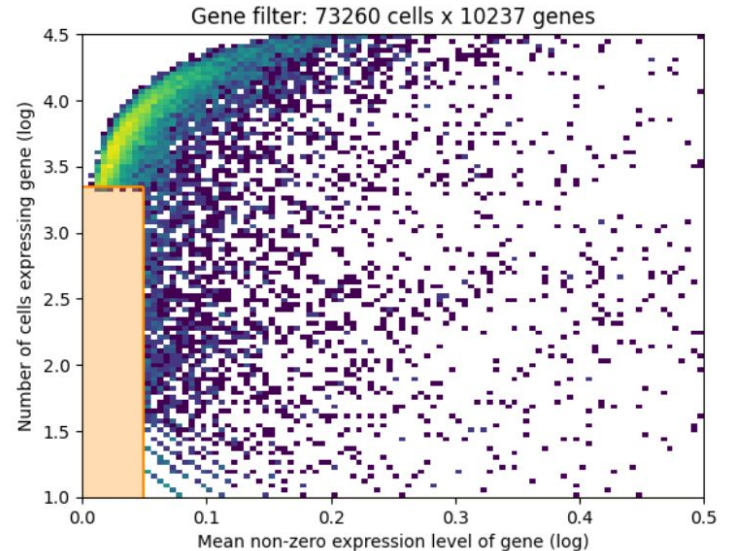
Filtering genes

Remove mitochondrial genes (gene names start with prefix mt- or MT-):

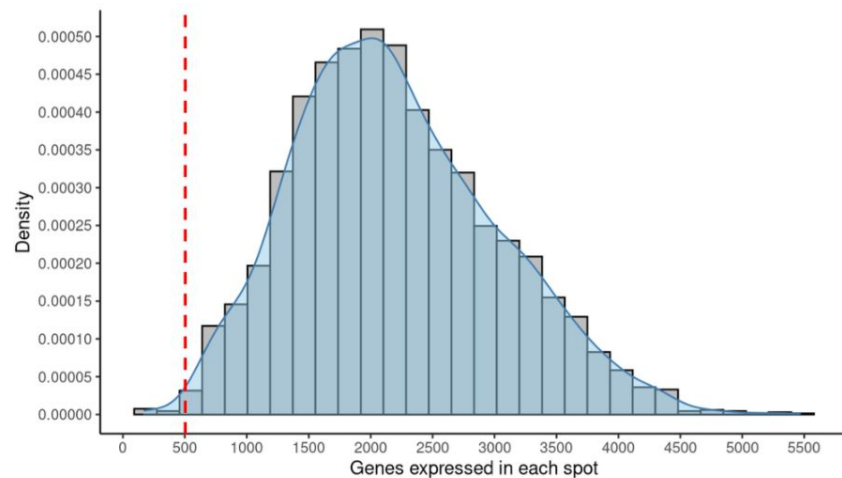
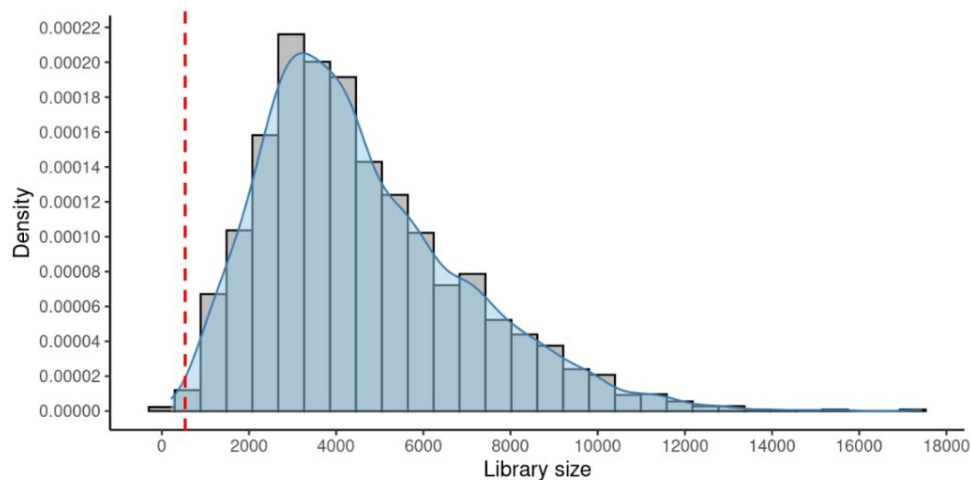
- They compose 15-40% of mRNA in each location
- Mostly representing technical artifacts

Possibly other artifacts, e.g. Hb for blood contamination etc.

Filter based on cell count, cell percentage, non-zero mean (aim at 8-16 k genes)

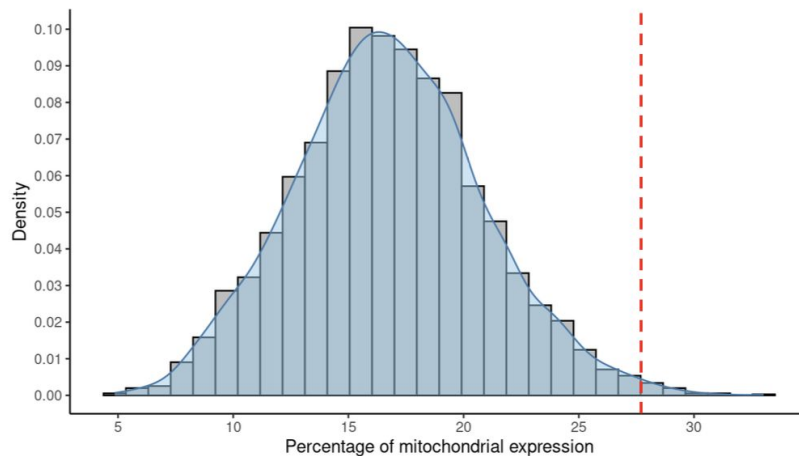


Filtering spots (counts / number of genes)



Poor mRNA capture rates due to cell damage and missing mRNAs, or low reaction efficiency.

Filtering spots (% mitochondrial)

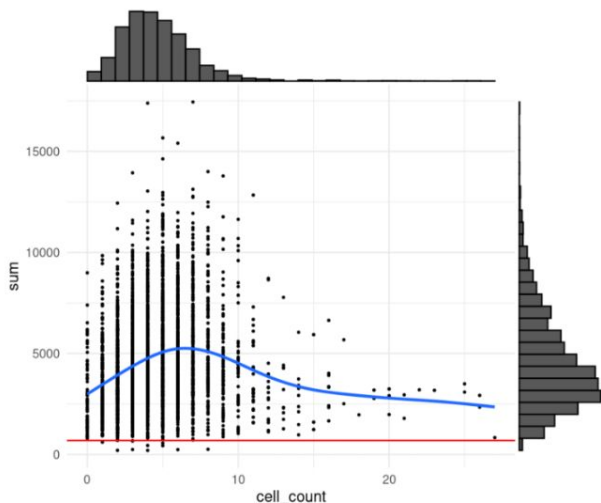


Partial cell lysis leading to leakage and missing cytoplasmic mRNAs, with the resulting reads therefore concentrated on the remaining mitochondrial mRNAs.

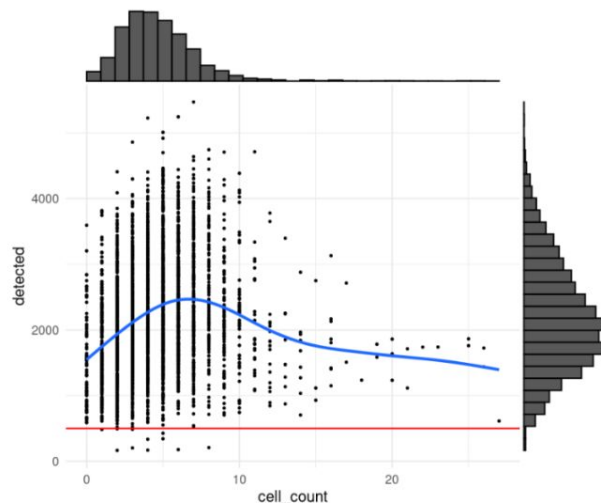
Spots may contain zero, one, or multiple cells

We can plot the library sizes against the number of cells per spot (to make sure we don't remove any spots that may have biological meaning)

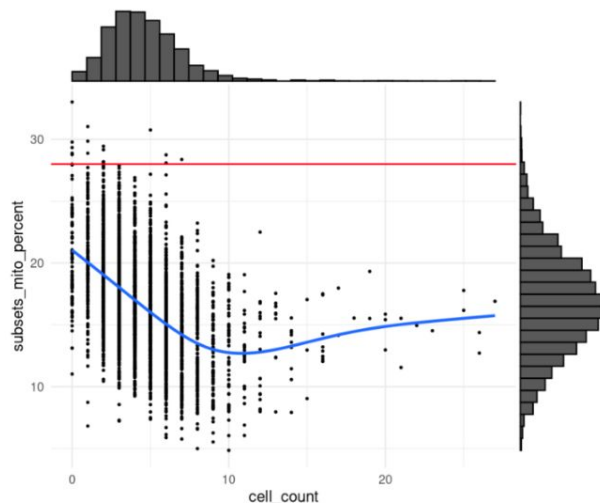
Library size



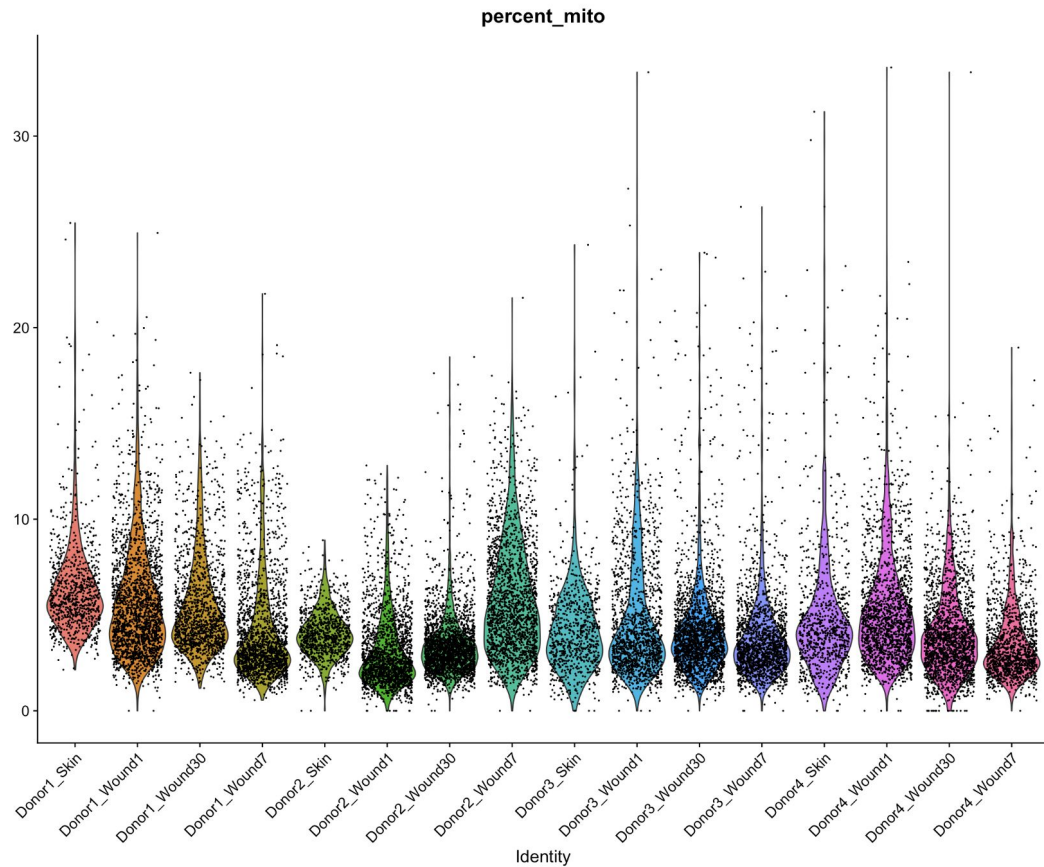
Total Genes



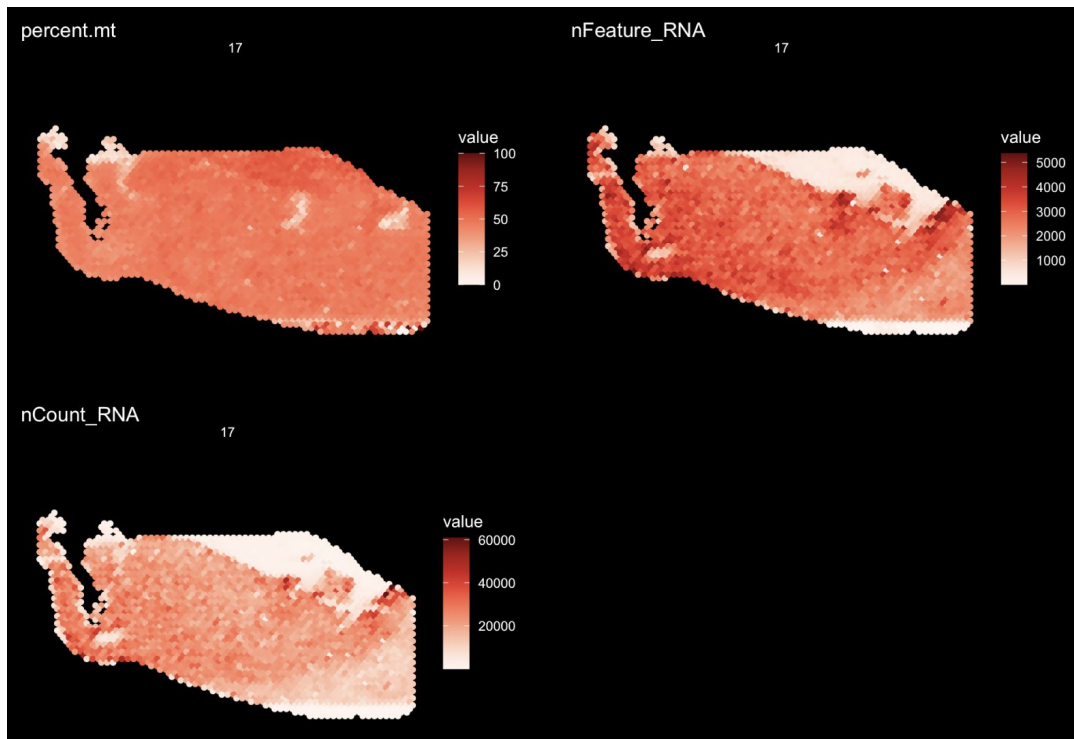
Mitochondrial %



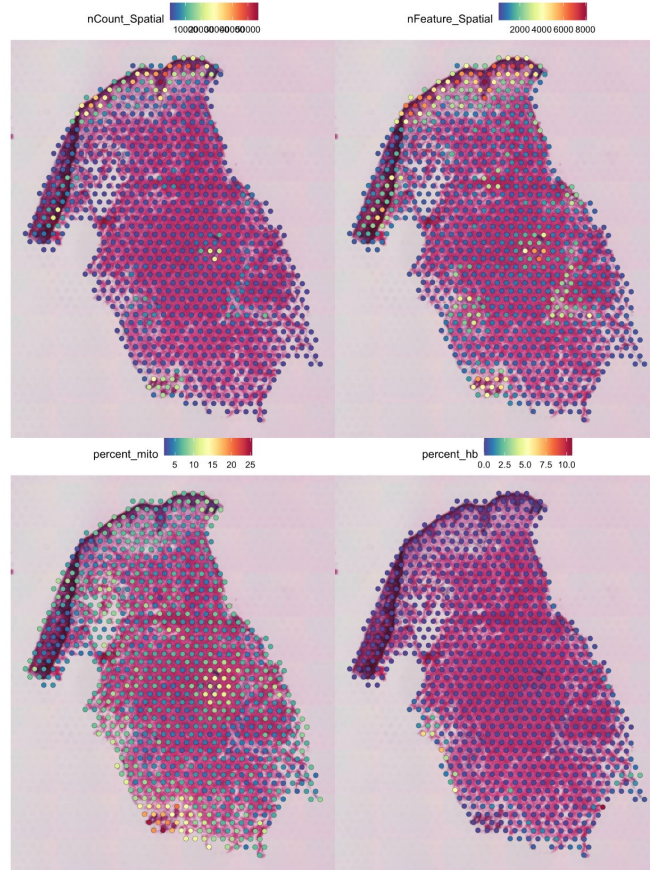
Always check individual sections



Check spatial patterns of QC

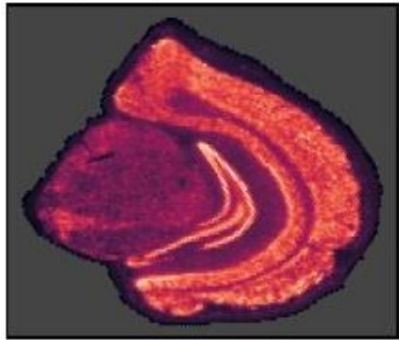


Quality vs biological variation



Higher cell density give larger library sizes

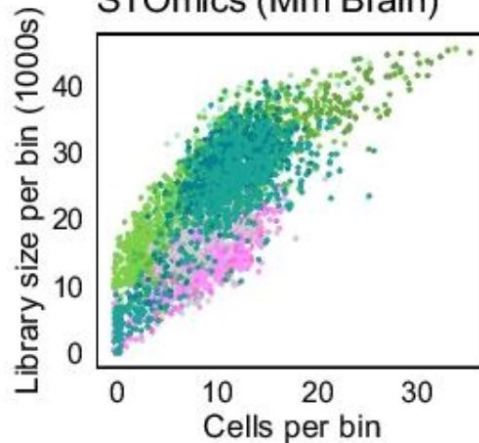
STOmics (Mm Brain)
Library size per bin: 0–26.5k



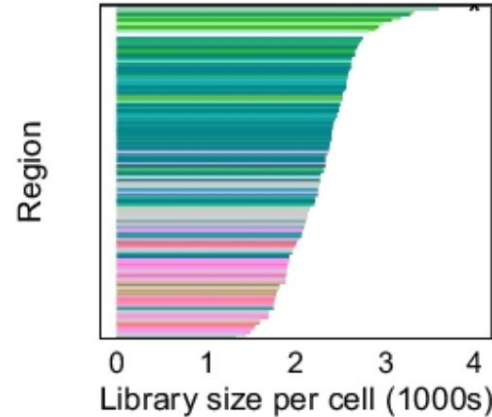
STOmics (Mm Brain)



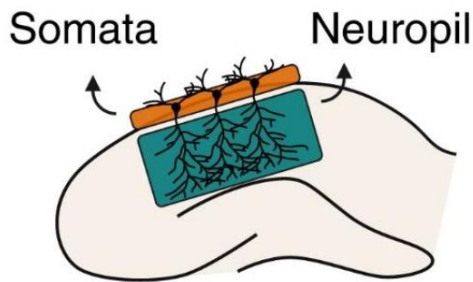
STOmics (Mm Brain)



STOmics (Mm Brain)



mRNA localization within cells



Neuropil comprises dendrites, axons, glia, and a sparse population of interneurons, but lacks principal neuron cell bodies

- sc/snRNA-seq:
 - mRNA transcripts from a cell body or nucleus
 - SRT:
 - mRNA transcripts from multiple cells?
 - Tissue architecture can result in differences in reagent permeability
- This would lead to sampling differences across the tissue and subsequently library size differences and QC differences

Cajigas*, Tushev*,... (2012)

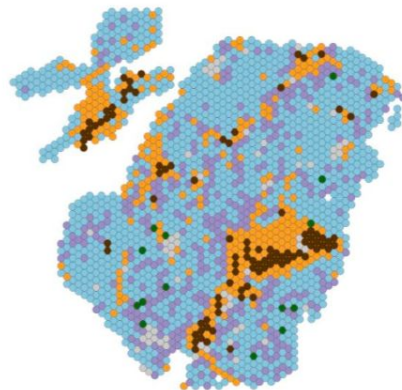
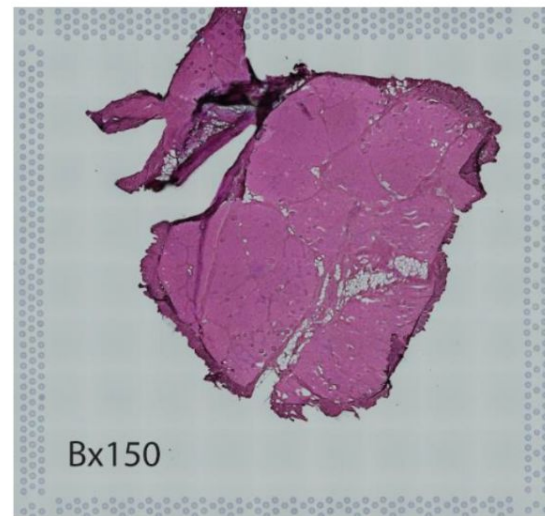
Quality vs biological variation

H&E

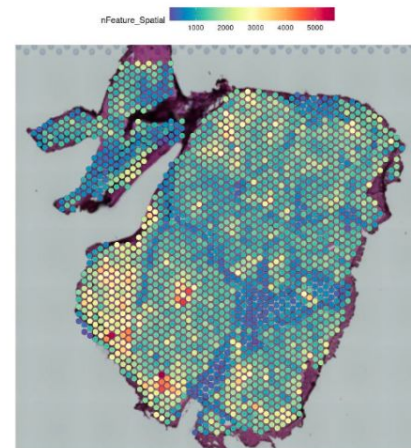
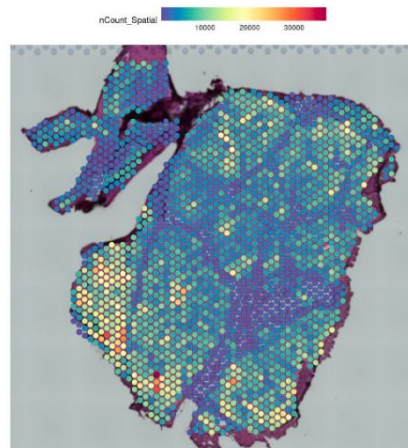
Annotation

Library size

Total Genes

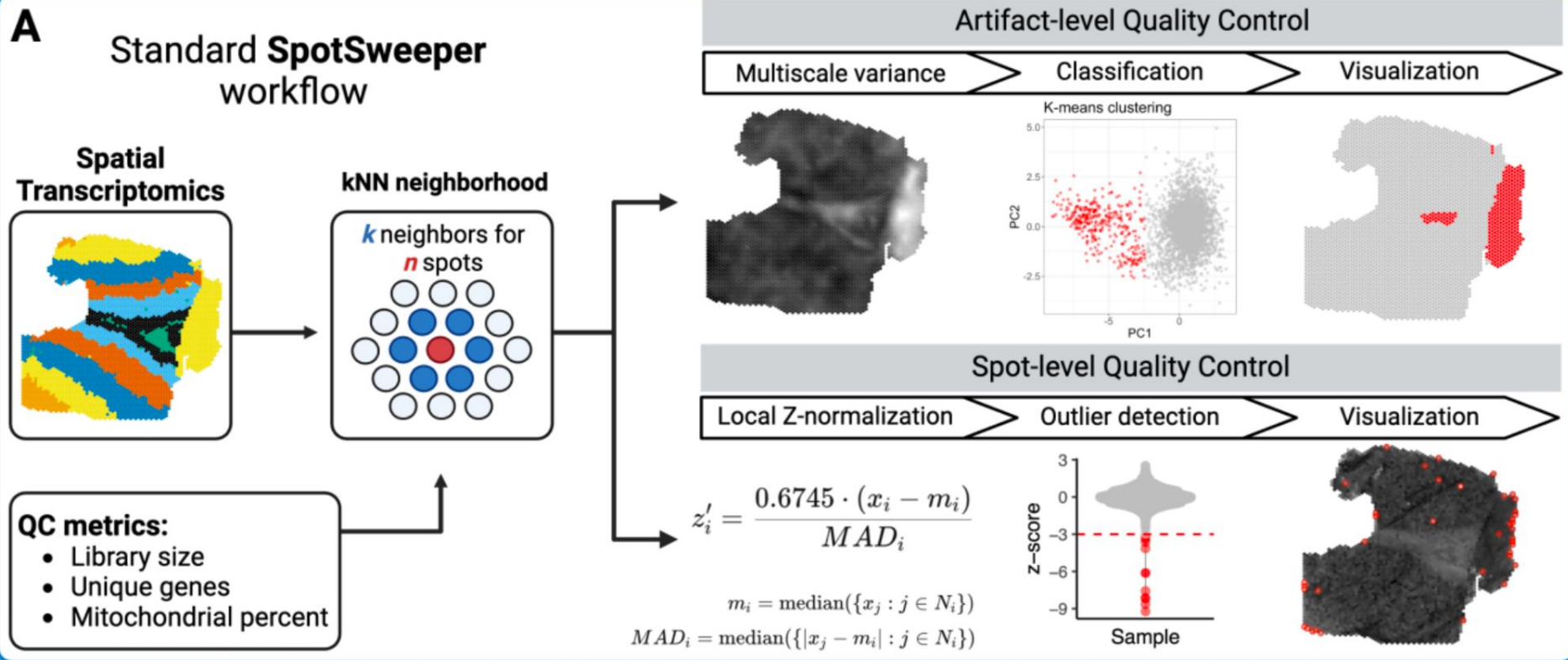


- Type 1
- Type 2A
- Type 2X
- Connective tissue
- Fat
- Non muscle fibers



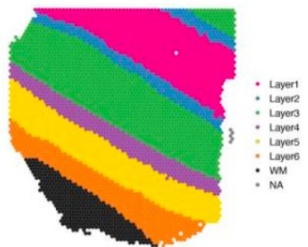
Heezen*, Mao*..., Mahfouz*, Spitali*(submitted)

Spatially-aware QC (SpotSweeper)

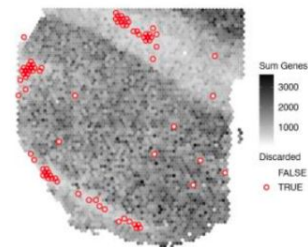


Spatially-aware QC (SpotSweeper)

Annotation

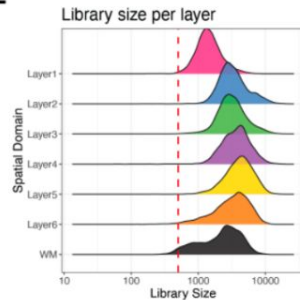


Global

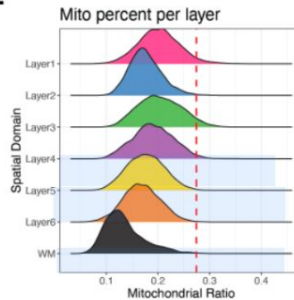


Global outliers

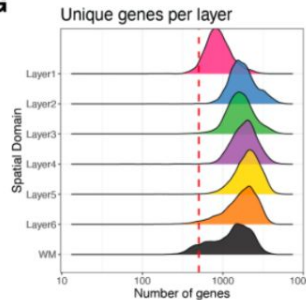
E



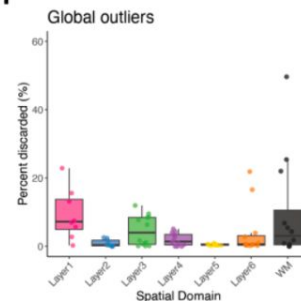
F



G

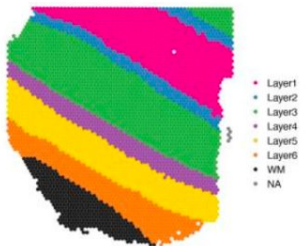


H

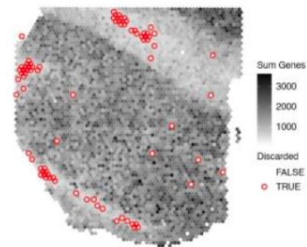


Spatially-aware QC (SpotSweeper)

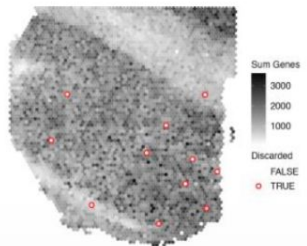
Annotation



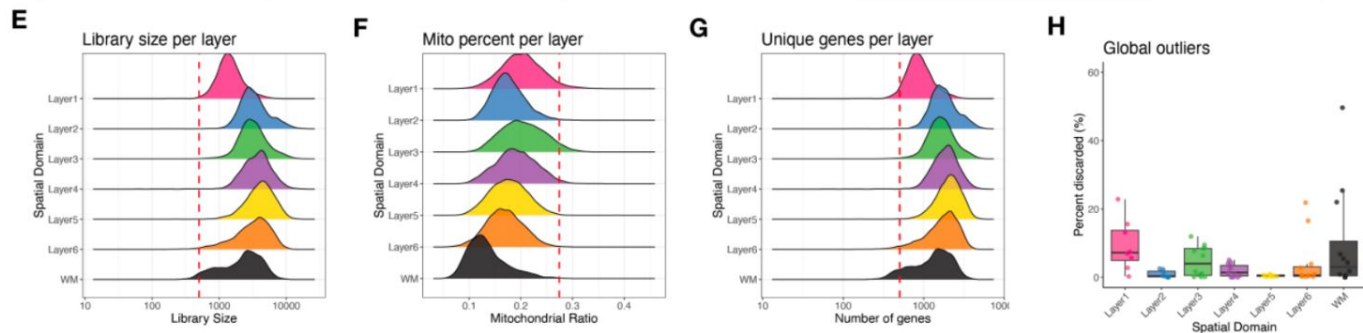
Global



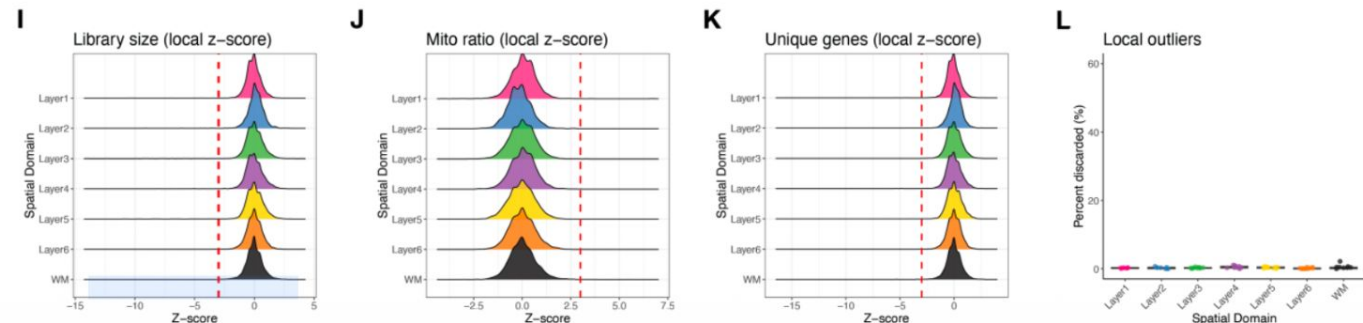
Local



Global outliers



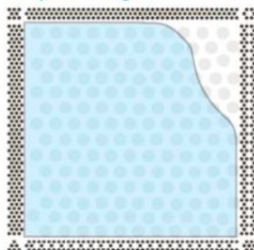
Local outliers



SpotSweeper (region-level artifacts)

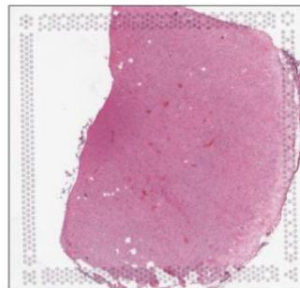
Incomplete coverage of Visium array

Liquid reagent

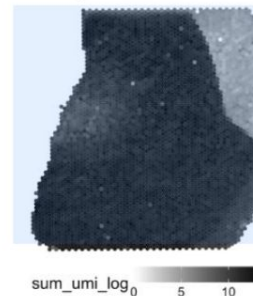


Dry spots result in smaller library size and fewer genes detected

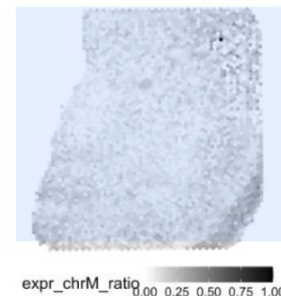
Br3942_mid



Library size



Mito Ratio



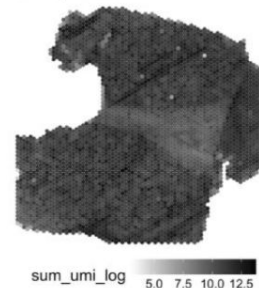
BayesSpace k=7



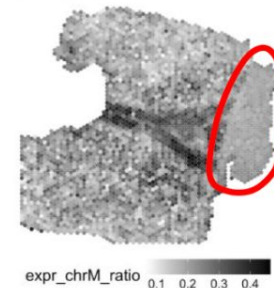
Br8325_ant



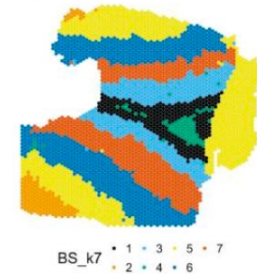
Library size



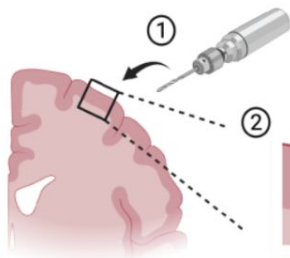
Mito Ratio



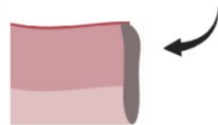
BayesSpace k=7



Tissue damage during dissection

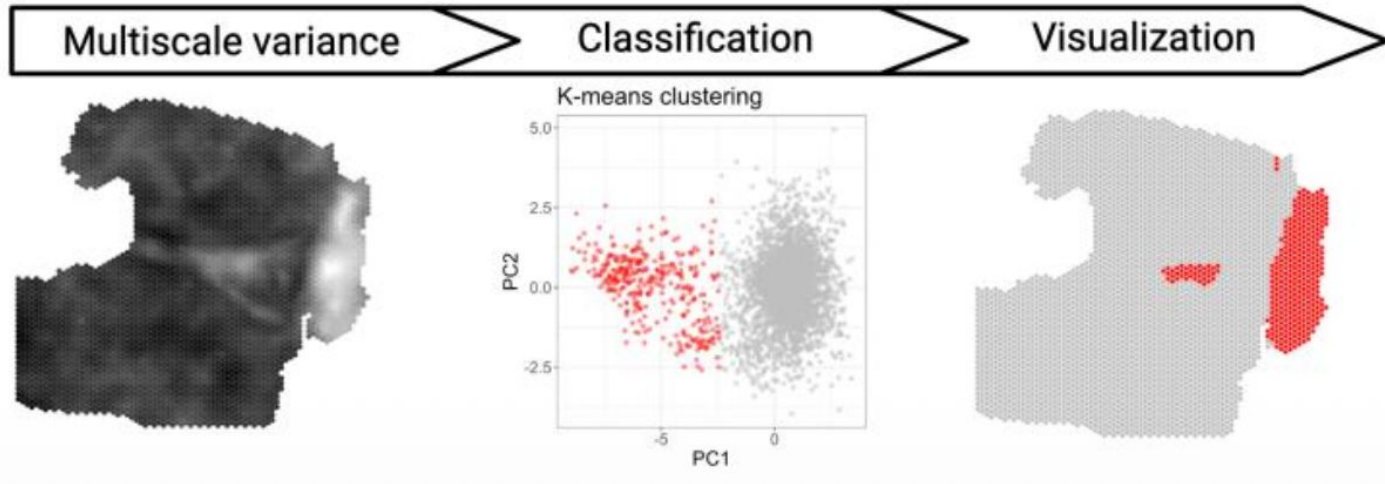


Hangnails from damaged tissue result in loss of biological signal



SpotSweeper (region-level artifacts)

1. The k-NN for each spot are identified based on the spatial coordinates
2. For each neighborhood size (i.e., scale), local variance of the mitochondrial ratio is calculated and adjusted for a mean-variance relationship using linear regression
3. Mean-corrected local variance = the residuals of the linear regression
4. Perform PCA on the mean-corrected local variances of all neighborhood sizes
5. Apply k-means clustering (k=2) in the first two PCs to identify regional artifacts compared to high-quality tissue



Quality Control - Summary

Filtering out low quality regions will help remove artifacts.

Always look at the QC data both as distributions and in the spatial context.

Be aware! Get to know the biology of your tissue to understand what you are removing.

Normalization

Normalization

- Biological variation:
 - Tissue type/state
 - Cellular composition
 - Cell size
 - Sex, Age, ...
 - Etc..
- Technical variation
 - Tissue quality
 - Library prep efficiency
 - Batch effects
 - Etc...

Normalization

- Biological variation:
 - Tissue type/state
 - Cellular composition
 - Cell size
 - Sex, Age, ...
 - Etc..
- Technical variation
 - Tissue quality
 - Library prep efficiency
 - Batch effects
 - Etc...

To identify cell or tissue types we would like to remove all other sources of variation.

Normalization

- Want to make expression comparable across samples, spots/cells and genes.
- Involves 3 main steps:
 - Scaling
 - Transformation
 - Removal of unwanted variation

Scaling Normalization

- **Depth normalization** – for uneven sequencing depth
- **Gene length normalization** – for differences in gene detection due to gene length (full length methods)
- **Drop-out rate normalization** – for differences in RNA content / drop-out rates
- **Cell area normalization** - size of segmented area

OBS! After scaling we have relative amounts of the different genes, not absolute values.

Depth Normalization

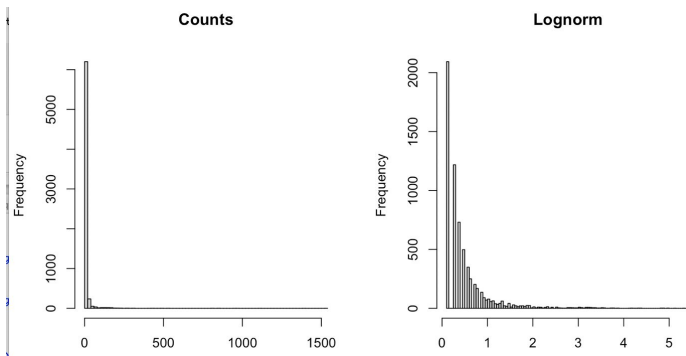
- In most cases the amount of RNA – and of UMIs/reads differ between regions.
 - Can be biological, different cell types have different RNA amount. Different tissue regions are more/less dense.
 - Can be technical, RT-efficiency may differ between regions.
- NOTE! Also important to check for outlier genes that constitute large proportion of the reads!

Transformation Normalization

- Idea is to have a distribution of expression and variance in expression values that best captures biological variation.
- Most common transformation - log-transform

Log-transformation

- Log-transformed values approaches normal distribution for bulk RNAseq data
- For scRNAseq – more similar to zero-inflated binomial
- Still more similar to normal distribution than raw counts.

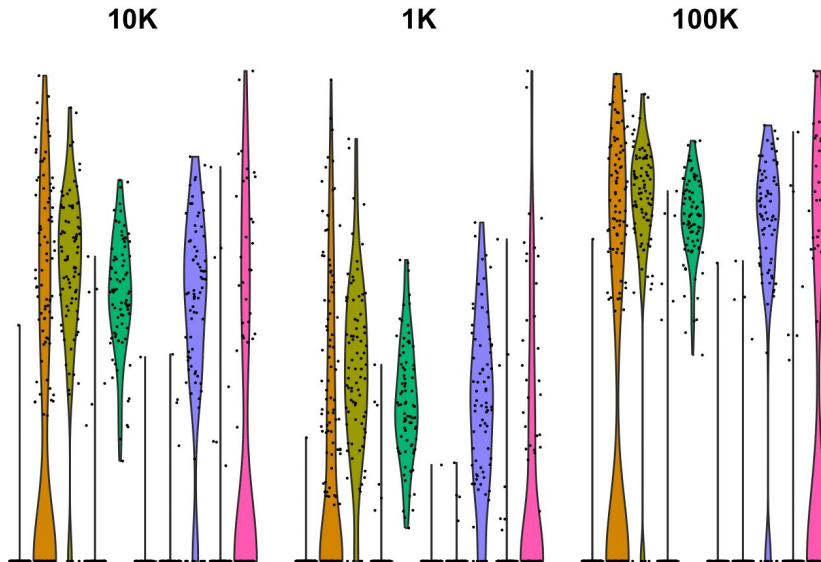


Scaling factors and pseudocounts

Default scale factor is often 10 000, and pseudocount 1.

Relative counts: $c/\text{sum}(c) * \text{scale.factor}$

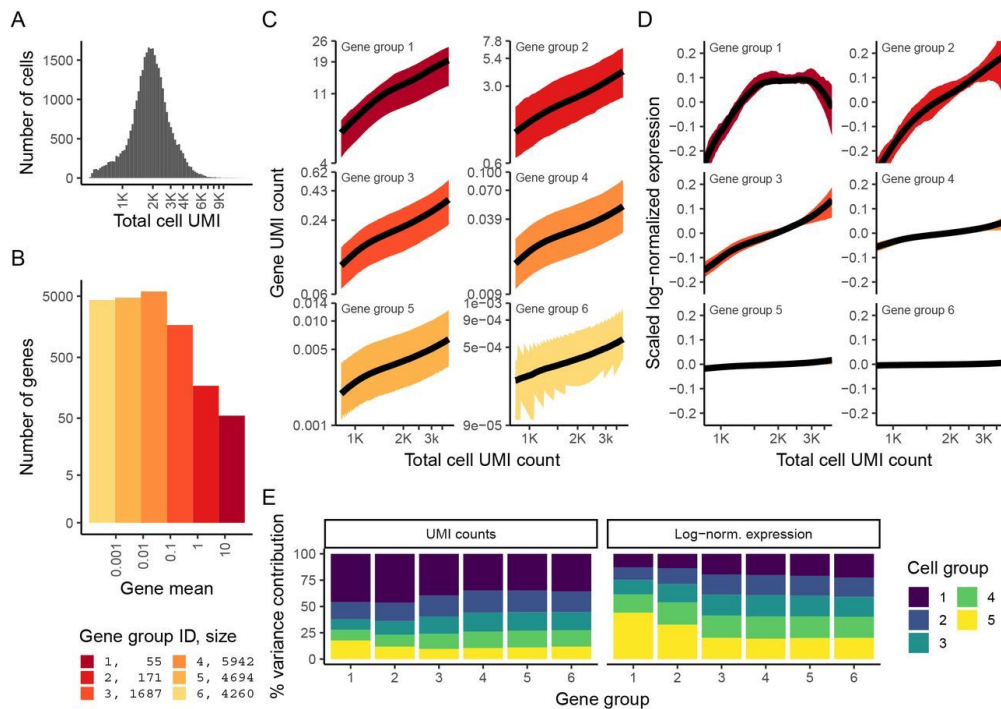
Lognorm: $\log(\text{RC}+1)$



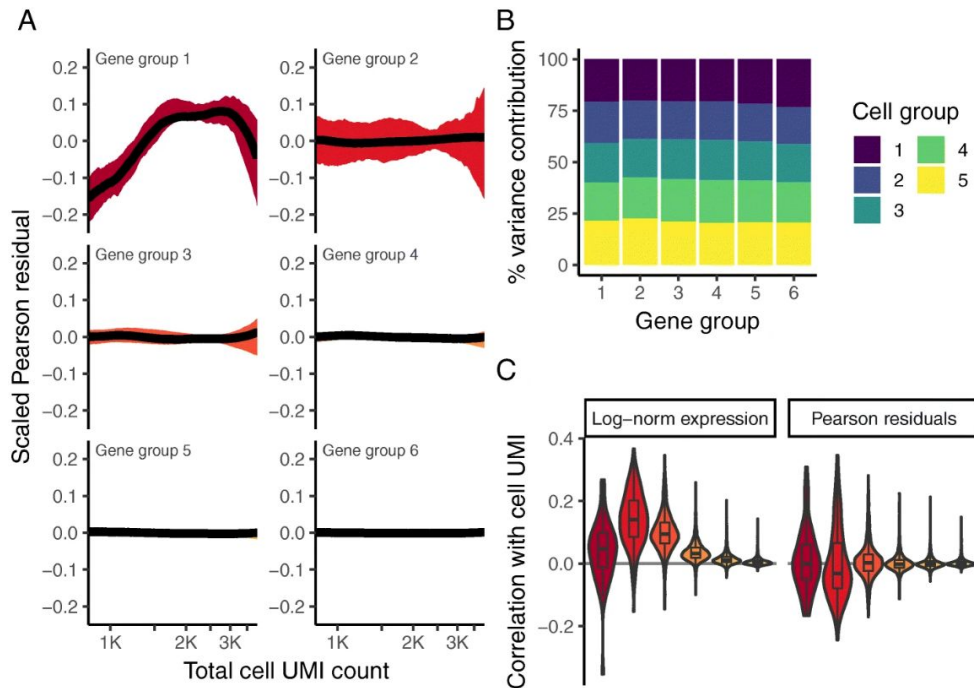
Normalization within gene groups

- Global scale factors may lead to overcorrection for weakly and moderately expressed genes and undercorrection for highly expressed genes.
- It will also differ a lot between cells with high/low total counts.
- Solution: Do normalization by regression to total counts – SCNorm & SCTransform

SCTransform



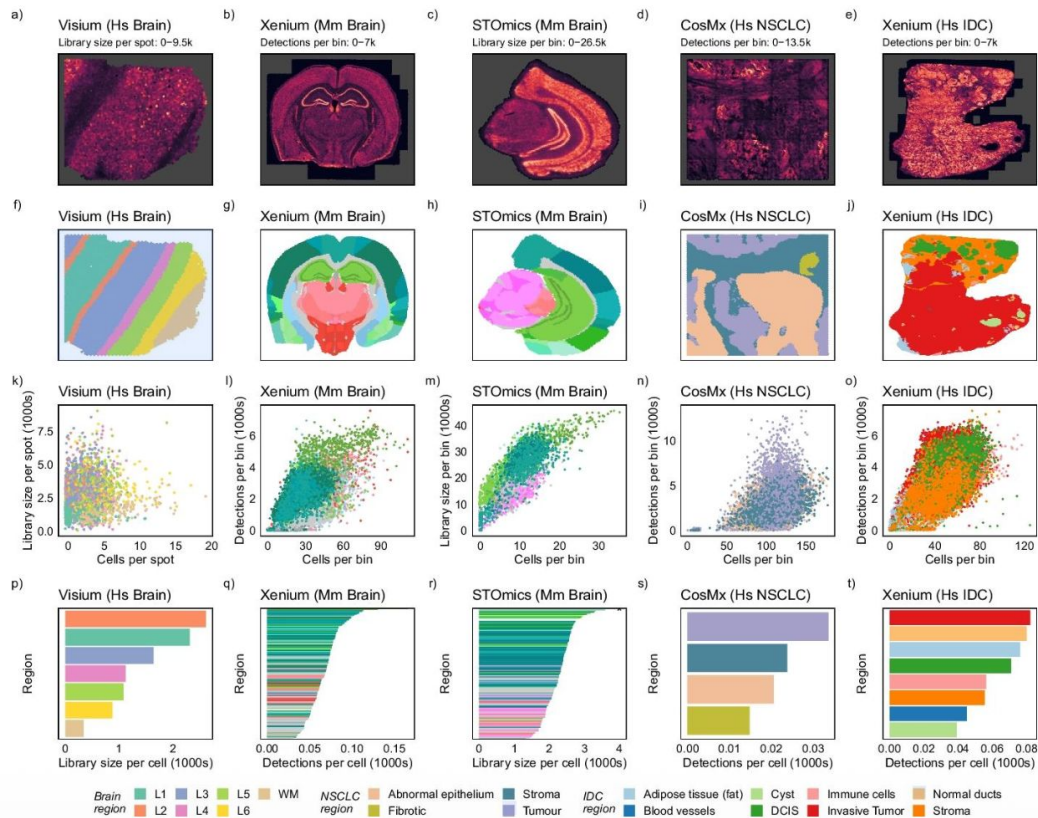
SCTransform



Pearson residuals from regularized negative binomial (NB) regression

Library size $\approx$ biological variation

Library size normalization assumes that all regions of the tissue have the same underlying mRNA abundance



Which normalization method to use?

Bhuva et al. *Genome Biology* (2024) 25:99
<https://doi.org/10.1186/s13059-024-03241-7>

Genome Biology

SHORT REPORT

Open Access

Library size confounds biology in spatial transcriptomics data



Dharmesh D. Bhuva^{1,2,3*} , Chin Wee Tan^{2,3,4}, Agus Salim^{2,5}, Claire Marceau^{3,6}, Marie A. Pickering⁷, Jinjin Chen^{2,3}, Malvika Kharbanda^{1,2,3}, Xinyi Jin^{2,3}, Ning Liu^{1,2,3}, Kristen Feher^{1,2,3}, Givanna Putri^{2,3}, Wayne D. Tilley⁷, Theresa E. Hickey⁷, Marie-Liesse Asselin-Labat^{3,6}, Belinda Phipson^{2,3†} and Melissa J. Davis^{1,2,3,4,8†}

- Tested the effects of normalization on spatial domain identification

Though sctransform removes library size effects effectively, their confounding with biology results in removal of biological effects as well.

See also for imaging-based SRT...

Atta et al. *Genome Biology* (2024) 25:153
<https://doi.org/10.1186/s13059-024-03303-w>

Genome Biology

RESEARCH

Open Access

Gene count normalization in single-cell imaging-based spatially resolved transcriptomics



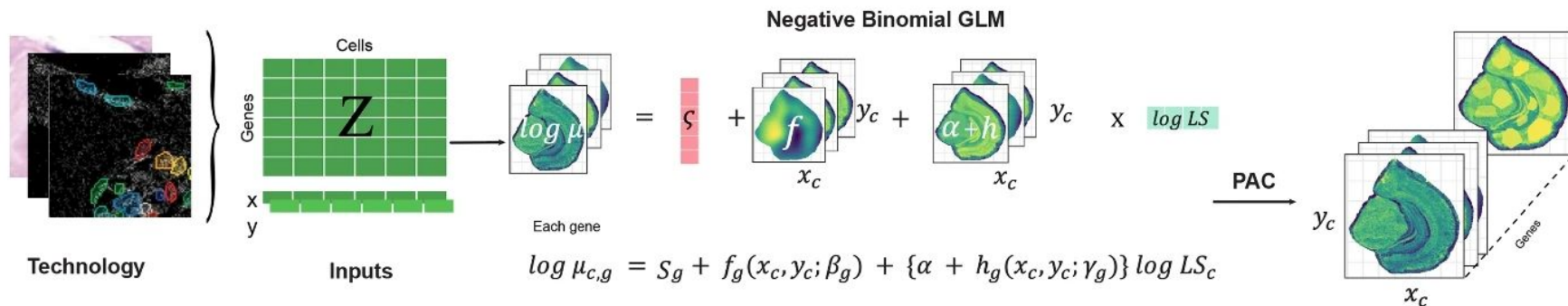
Lyla Atta^{1,2}, Kalen Clifton^{1,2}, Manjari Anant^{2,3}, Gohta Aihara^{1,2} and Jean Fan^{1,2*} 

SpaNorm

Optimally decomposing spatially-smooth variation into library size associated and library size independent variation via generalized linear model (GLM)

Computing spatially smooth location- and gene-specific scaling factors

Using percentile-invariant adjusted counts (PAC) as normalized data for downstream analyses.



Feature (Gene) selection

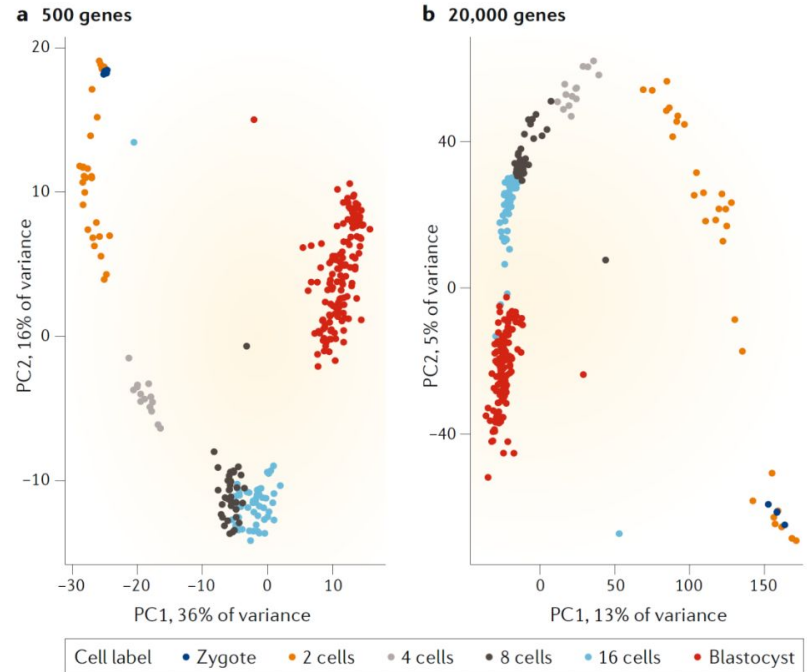
Feature selection

Curse of dimensionality:

- More features (genes) -> noise dominates distances between samples (cells), effectively all cells get 'same' distance

Remove genes which only exhibit technical noise

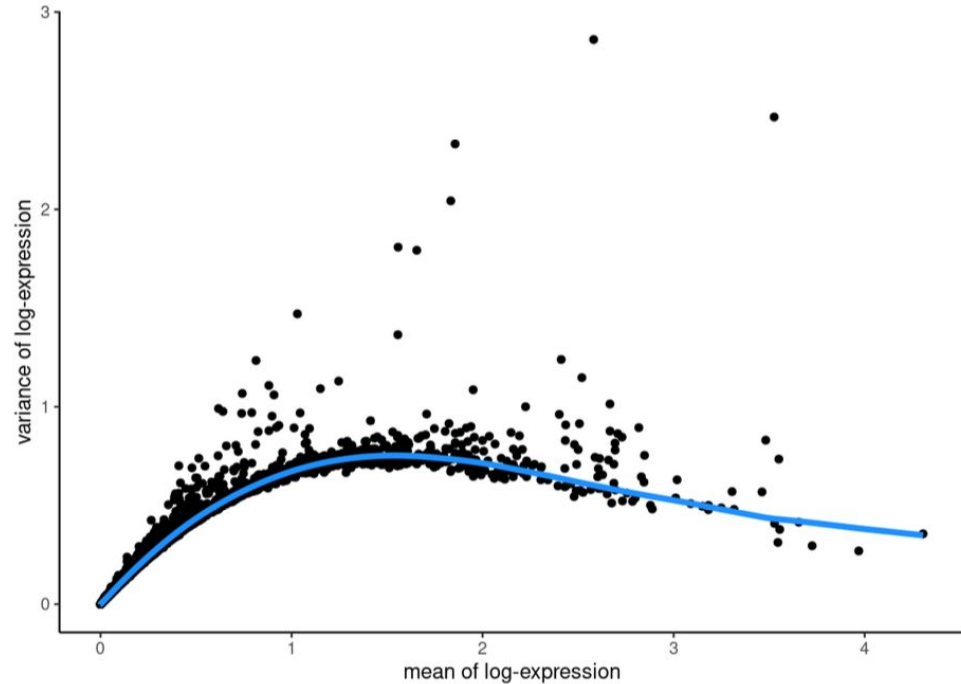
- Increase the signal:noise ratio
- Reduce the computational complexity



Feature selection (HVG)

Highly Variable Genes (HVG)

- Quantifying **per-gene variation** as the variance of the log-normalized expression values of each gene
- We define a **technical component** based on the average
- We define the **biological component** as the difference between its total variance and the technical component.



Feature selection (HVG)

How many genes should you chose?

Too few variable genes:

- Do not capture all biological signal.

Too many variable genes:

- Risk of also including technical noise.
- Higher computational cost.

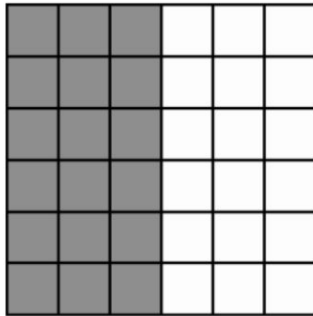
No number fits all datasets!

- Single cell type / homogeneous tissue - use few HVGs (1000-2000)
- Many cell types / complex tissue - use more HVGs (3000-5000)

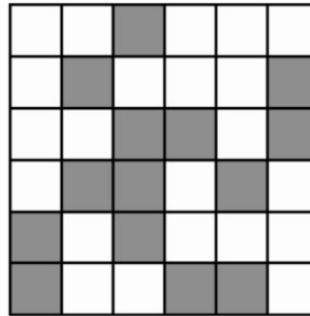


Feature selection (Spatially variable genes)

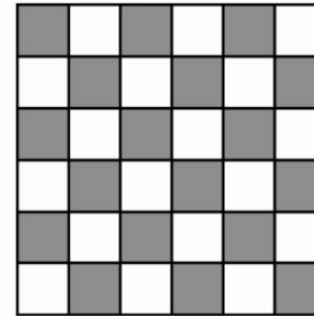
"The first law of geography: Everything is related to everything else, but near things are more related than distant things." Waldo R. Tobler ([Tobler 1970](#))



Positive spatial
autocorrelation



No spatial
autocorrelation



Negative spatial
autocorrelation

Feature selection (Spatially variable genes)

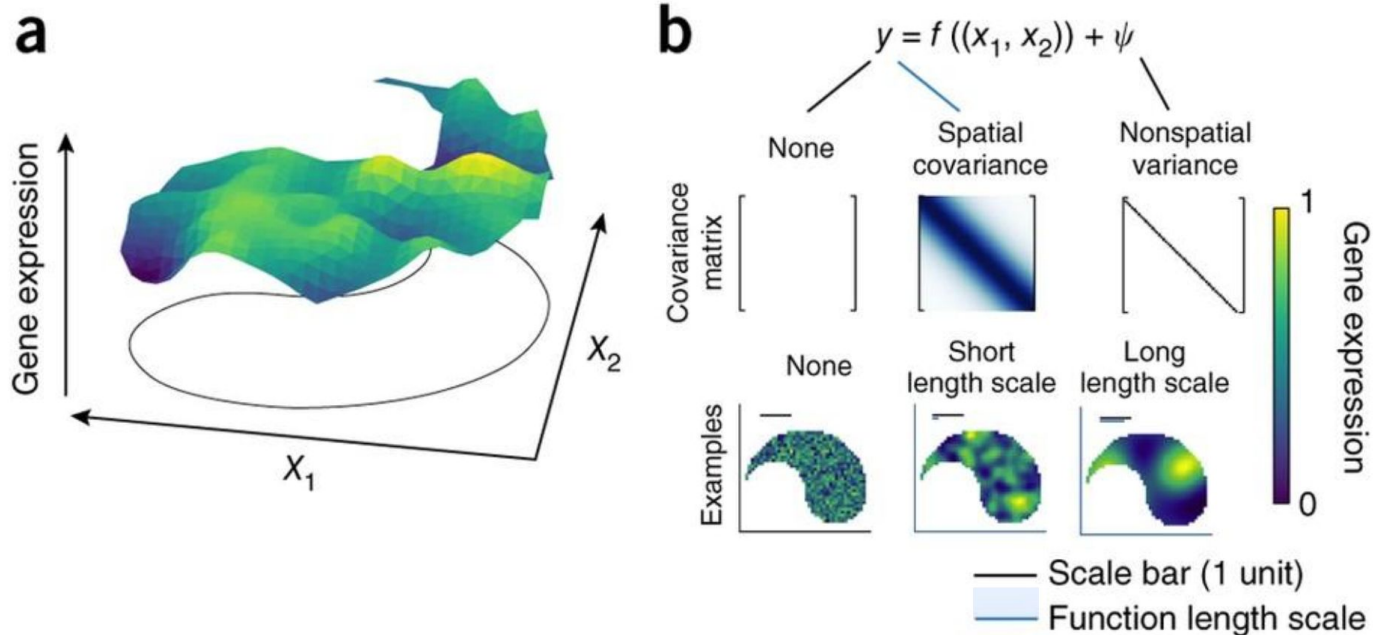
- SVGs are genes with a highly spatially correlated pattern of expression, which varies along with the spatial distribution of a tissue structure of interest
- Statistical measures of spatial autocorrelation:
 - Moran's I
 - Geary's
 -

Feature selection (Spatially variable genes)

SpatialDE: Svensson et al. Nature Methods 2018

SPARK: Sun et al. Nature Methods 2020

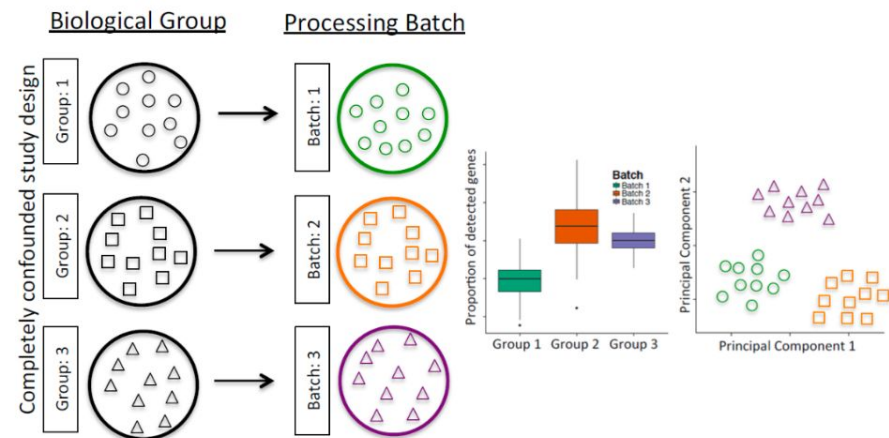
SPARK-X: Zhu et al. genome Biology 2021



Batch correction

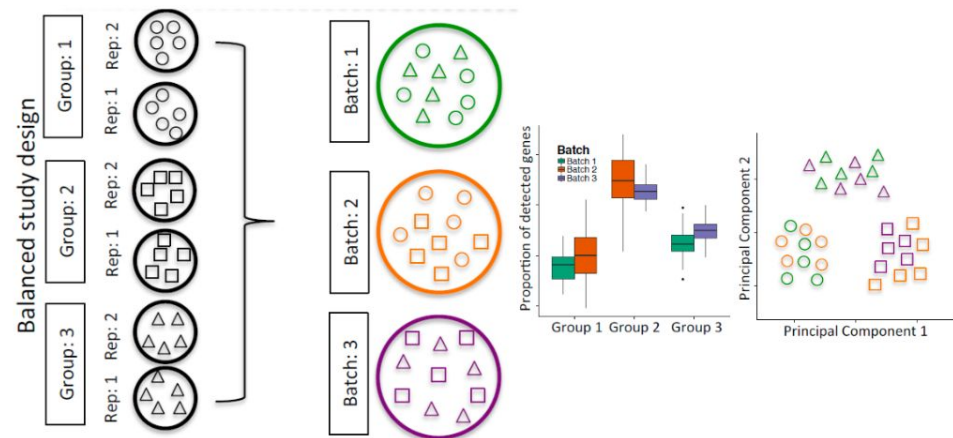
Batch correction

Confounded design



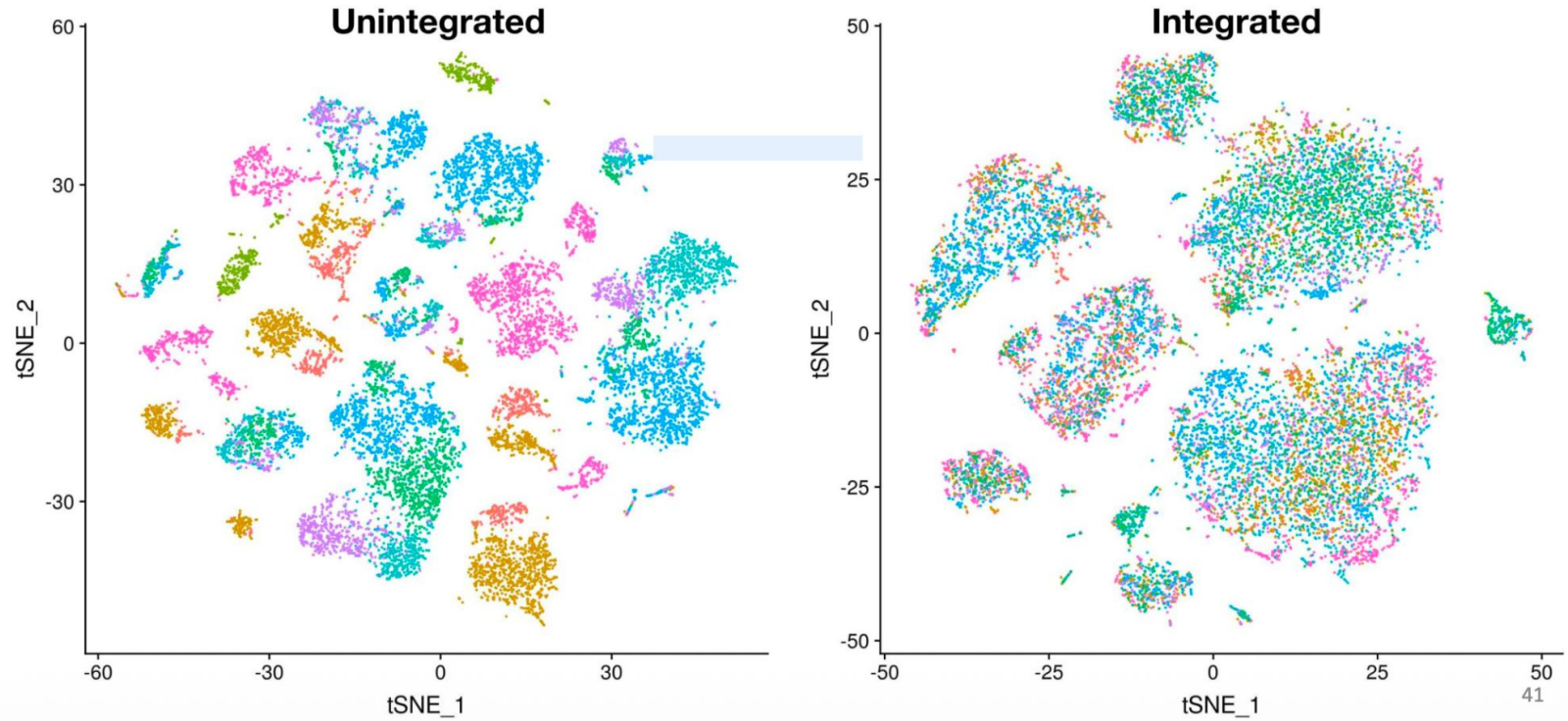
Don't design your experiment like this!!!

Not confounded design



Good experimental design *does not remove batch effects*, it prevents them from biasing your results.

Batch correction



Batch correction methods from scRNAseq

- scVI (<https://doi.org/10.1038/s41592-018-0229-2>)
- MNNcorrect (<https://doi.org/10.1038/nbt.4091>)
- CCA + anchors (Seurat v3) (<https://doi.org/10.1101/460147>)
- CCA + dynamic time warping (Seurat v2) (<https://doi.org/10.1038/nbt.4096>)
- LIGER (<https://doi.org/10.1101/459891>)
- Harmony (<https://doi.org/10.1101/461954>)
- Conos (<https://doi.org/10.1101/460246>)
- Scanorama (<https://doi.org/10.1101/371179>)
- scMerge (<https://doi.org/10.1073/pnas.1820006116>)

Spatially aware batch correction

Precast (<https://doi.org/10.1038/s41467-023-35947-w>)

STAlinger (<https://doi.org/10.1038/s43588-023-00528-w>)

GraphST (<https://doi.org/10.1038/s41467-023-36796-3>)

STitch3D (<https://doi.org/10.1038/s42256-023-00734-1>)

SpaGCN (<https://doi.org/10.1038/s41592-021-01255-8>)

STAGATE (<https://doi.org/10.1038/s41467-022-29439-6>)

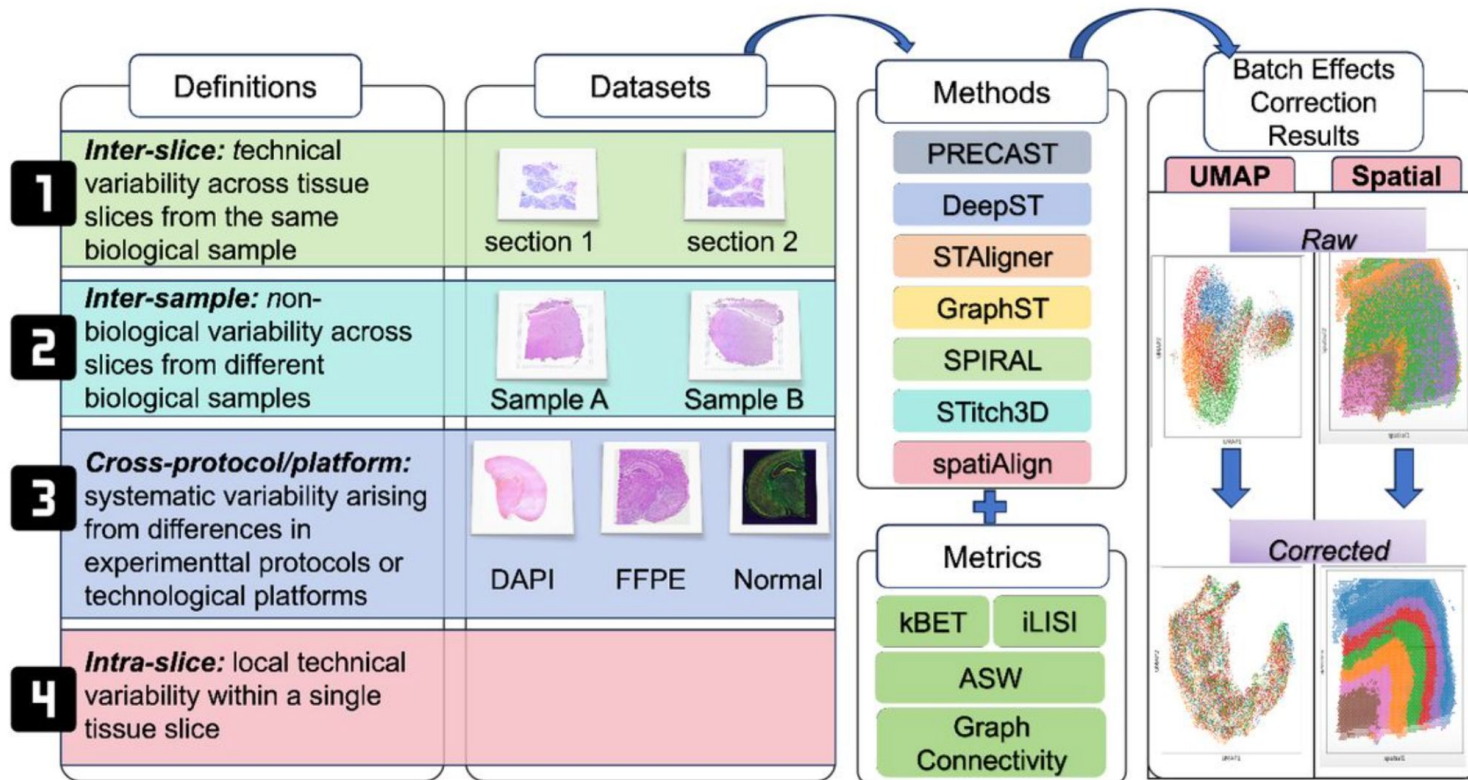
PASTE / PASTE2 (<https://doi.org/10.1101/gr.277670.123>)

SPIRAL (<https://doi.org/10.1186/s13059-023-03078-6>)

DeepST (<https://doi.org/10.1093/nar/gkac901>)

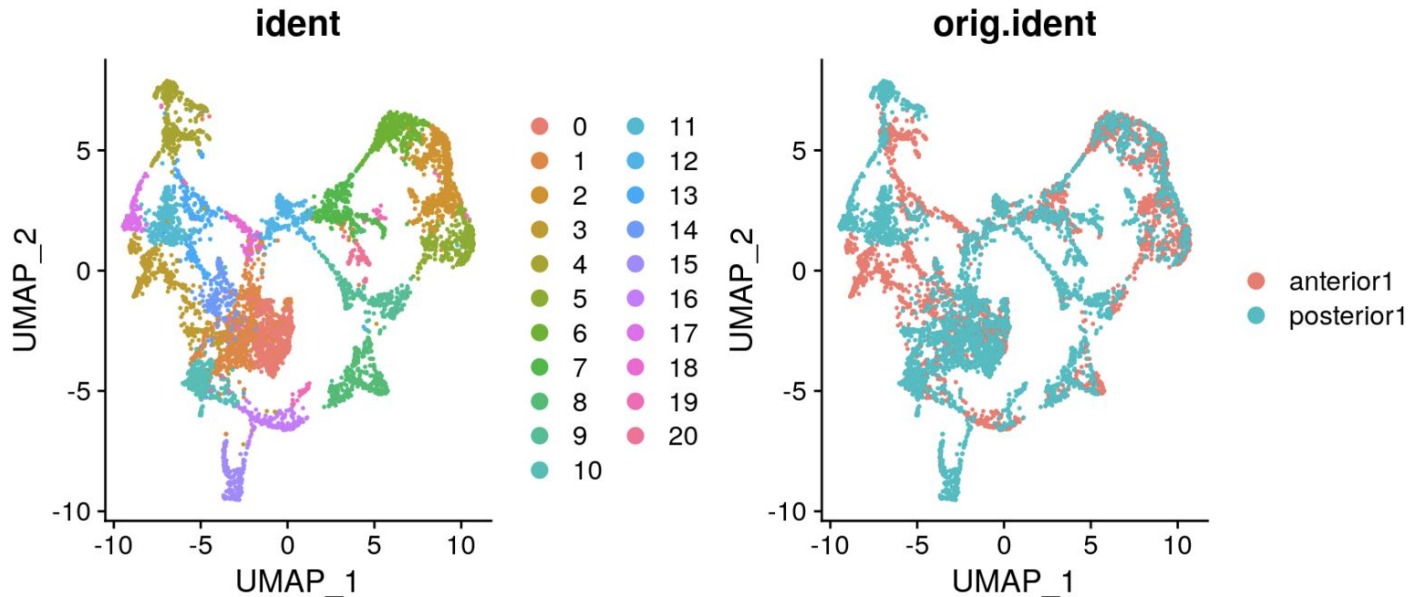
INSTINCT (<http://dx.doi.org/10.1038/s41467-025-56535-0>)

Spatially aware batch correction



Dimensionality reduction

- PCA is performed as a first step to reduce dimensions and remove technical noise.
 - e.g. first 10-50 PCs are selected for further analysis.
- UMAP - Uniform Manifold Approximation and Projection - is commonly used in scRNAseq to visualise complex data in 2D



Dimensionality reduction

Many methods include a **spatial component**:

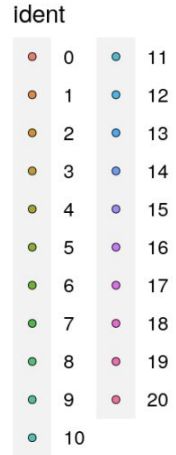
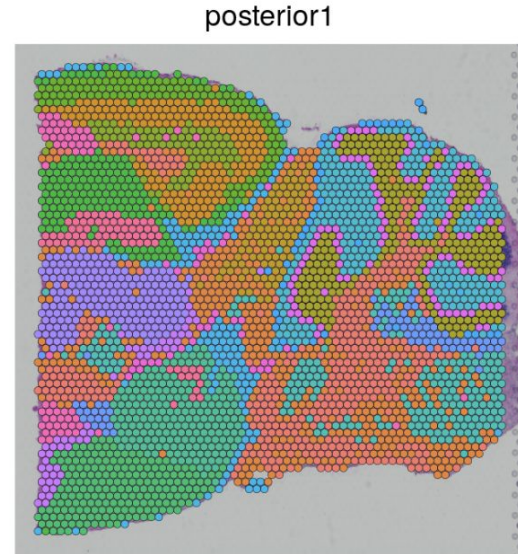
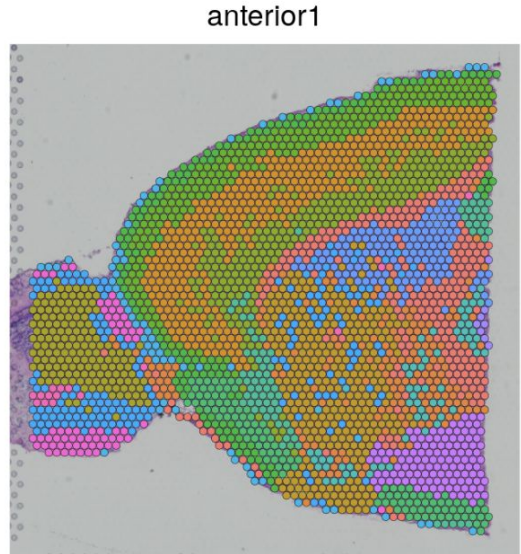
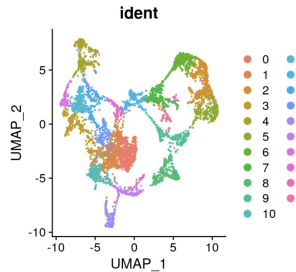
- SpatialPCA - <https://www.nature.com/articles/s41467-022-34879-1>
- Spatial Transcriptomics Analysis with topic Modeling to uncover spatial Patterns (STAMP) -
<https://www.nature.com/articles/s41592-024-02463-8>
- SpaHDmap - <https://www.nature.com/articles/s41556-025-01838-z>
- Randomized Spatial PCA (RASP) -
<https://doi.org/10.21203/rs.3.rs-6050441/v1>

Clustering

Community detection in KNN (K-nearest neighbors) and/or SNN (Shared Nearest Neighbors) **graphs** describing similarities between cells are used.

- Most common algorithm is **Leiden** clustering

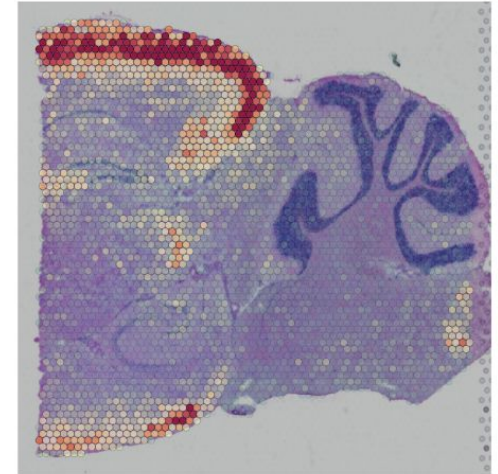
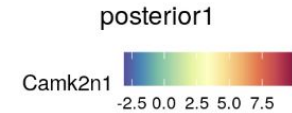
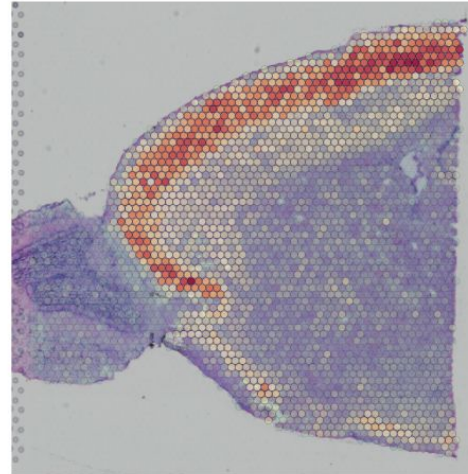
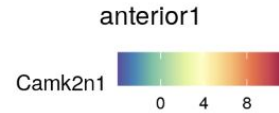
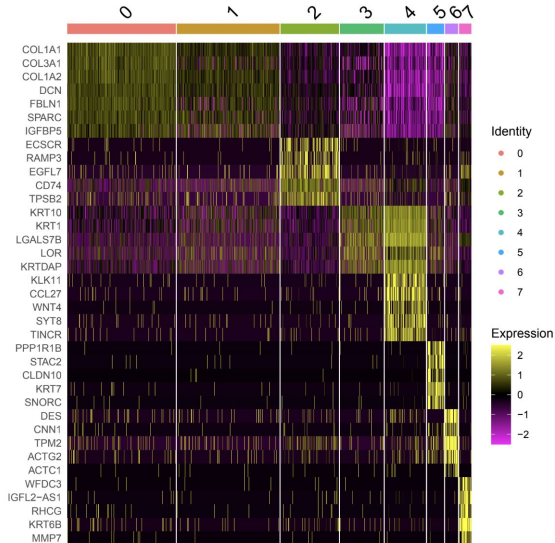
Niche or domain analysis can be seen as “clustering for SRT”



Differential expression

Once we have spatial clusters, we can run the same kind of differential expression tests as we do in scRNAseq to identify marker genes for the different regions

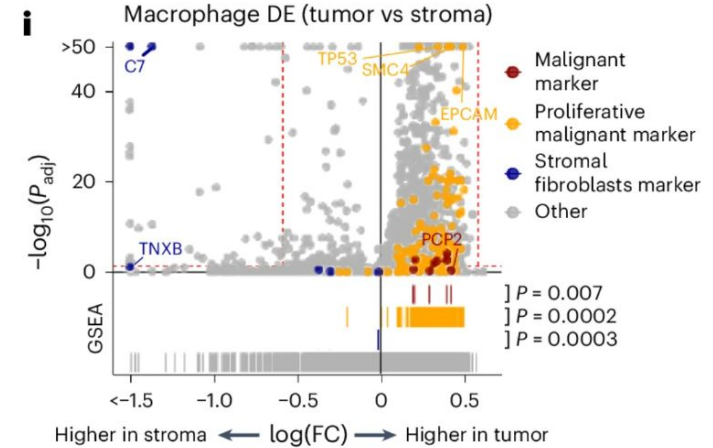
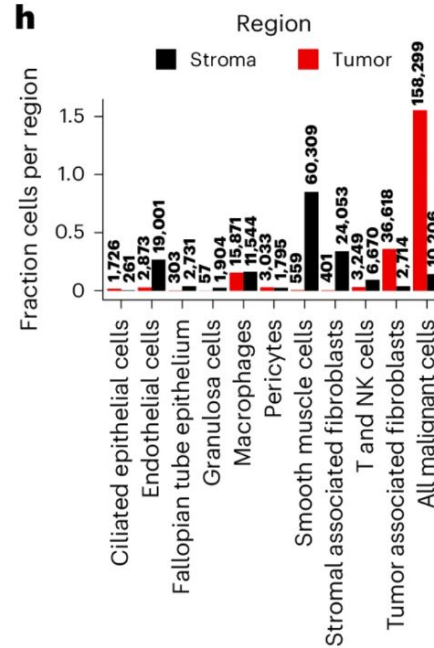
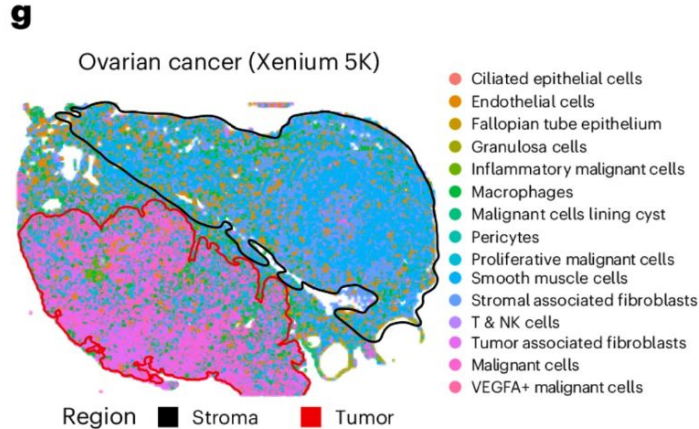
- Wilcoxon
- MAST
- t-Test
- Etc....



Differential expression

OBS! Be aware of the admixture effect:

- The signal is often confounded by surrounding cells

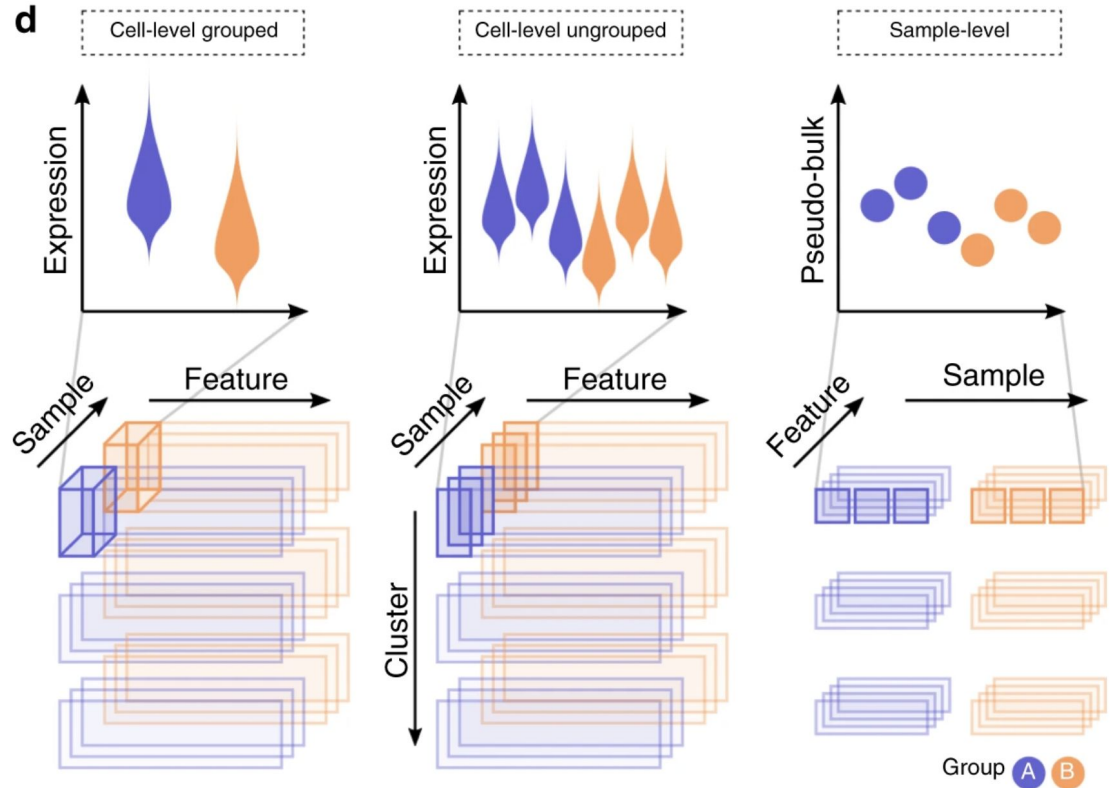


Differential expression

Multi-sample batch effects

Solutions:

- A model that includes a **random effect**
- **Pseudobulking**



Celltype prediction

Manual annotation based on marker genes per cluster

Automated annotation with similar methods to scRNAseq:

- Need to have a good scRNAseq/snRNAseq reference from the same tissue.
- Be aware that some celltypes may be lost during dissociation for scRNAseq

Celltypist - train the model on scRNAseq and predict a score for the spatial spots/cells

RCTD - train the model on scRNAseq and predict a score for the spatial spots/cells. Also gives a prediction of admixture of celltype signatures.

Summary

- Quality control (QC)
- Normalization
- Feature selection
- Batch correction
- Dimensionality Reduction
- Clustering
- Differential expression and annotation

Questions?